

Chapter 1

Axioms of Probability

1.1 INTRODUCTION

In search of natural laws that govern a phenomenon, science often faces “events” that may or may not occur. The event of *disintegration of a given atom of radium* is one such example because, in any given time interval, such an atom may or may not disintegrate. The event of *finding no defect during inspection of a microwave oven* is another example, since an inspector may or may not find defects in the microwave oven. The event that an *orbital satellite in space is at a certain position* is a third example. In any experiment, an event that may or may not occur is called **random**. If the occurrence of an event is inevitable, it is called **certain**, and if it can never occur, it is called **impossible**. For example, the event that an object travels faster than light is impossible, and the event that in a thunderstorm flashes of lightning precede any thunder echoes is certain.

Knowing that an event is random determines only that the existing conditions under which the experiment is being performed do not guarantee its occurrence. Therefore, the knowledge obtained from randomness itself is hardly decisive. It is highly desirable to determine quantitatively the exact value, or an estimate, of the chance of the occurrence of a random event. The theory of probability has emerged from attempts to deal with this problem. In many different fields of science and technology, it has been observed that, under a long series of experiments, the proportion of the time that an event occurs may appear to approach a constant. It is these constants that probability theory (and statistics) aims at predicting and describing as quantitative measures of the chance of occurrence of events. For example, if a fair coin is tossed repeatedly, the proportion of the heads approaches $1/2$. Hence probability theory postulates that the number $1/2$ be assigned to the event of *getting heads in a toss of a fair coin*.

Historically, from the dawn of civilization, humans have been interested in games of chance and gambling. However, the advent of probability as a mathematical discipline is relatively recent. Ancient Egyptians, about 3500 B.C., were using astragali, a four-sided die-shaped bone found in the heels of some animals, to play a game now called *hounds and jackals*. The ordinary six-sided die was made about 1600 B.C. and since then has been used in all kinds of games. The ordinary deck of playing cards, probably the most popular tool in games and gambling, is much more recent than dice. Although it is not known where and when dice originated, there are reasons to believe that they were invented in China sometime between the seventh and tenth centuries. Clearly, through gambling and games of chance people have gained intuitive ideas about the frequency of occurrence of certain events and, hence, about probabilities. But surprisingly, studies of the chances of events were not begun until the fifteenth century. The Italian scholars Luca Paccioli (1445–1514), Niccolò Tartaglia (1499–1557), Girolamo Cardano (1501–1576),

and especially Galileo Galilei (1564–1642) were among the first prominent mathematicians who calculated probabilities concerning many different games of chance. They also tried to construct a mathematical foundation for probability. Cardano even published a handbook on gambling, with sections discussing methods of cheating. Nevertheless, real progress started in France in 1654, when Blaise Pascal (1623–1662) and Pierre de Fermat (1601–1665) exchanged several letters in which they discussed general methods for the calculation of probabilities. In 1655, the Dutch scholar Christian Huygens (1629–1695) joined them. In 1657 Huygens published the first book on probability, *De Ratiocinates in Aleae Ludo* (*On Calculations in Games of Chance*). This book marked the birth of probability. Scholars who read it realized that they had encountered an important theory. Discussions of solved and unsolved problems and these new ideas generated readers interested in this challenging new field.

After the work of Pascal, Fermat, and Huygens, the book written by James Bernoulli (1654–1705) and published in 1713 and that by Abraham de Moivre (1667–1754) in 1730 were major breakthroughs. In the eighteenth century, studies by Pierre-Simon Laplace (1749–1827), Siméon Denis Poisson (1781–1840), and Karl Friedrich Gauss (1777–1855) expanded the growth of probability and its applications very rapidly and in many different directions. In the nineteenth century, prominent Russian mathematicians Pafnuty Chebyshev (1821–1894), Andrei Markov (1856–1922), and Aleksandr Lyapunov (1857–1918) advanced the works of Laplace, De Moivre, and Bernoulli considerably. By the early twentieth century, probability was already a developed theory, but its foundation was not firm. A major goal was to put it on firm mathematical grounds. Until then, among other interpretations perhaps the **relative frequency interpretation** of probability was the most satisfactory. According to this interpretation, to define p , the probability of the occurrence of an event A of an experiment, we study a series of sequential or simultaneous performances of the experiment and observe that the proportion of times that A occurs approaches a constant. Then we count $n(A)$, the number of times that A occurs during n performances of the experiment, and we define $p = \lim_{n \rightarrow \infty} n(A)/n$. This definition is mathematically problematic and cannot be the basis of a rigorous probability theory. Some of the difficulties that this definition creates are as follows:

1. In practice, $\lim_{n \rightarrow \infty} n(A)/n$ cannot be computed since it is impossible to repeat an experiment infinitely many times. Moreover, if for a large n , $n(A)/n$ is taken as an approximation for the probability of A , there is no way to analyze the error.
2. There is no reason to believe that the limit of $n(A)/n$, as $n \rightarrow \infty$, exists. Also, if the existence of this limit is accepted as an axiom, many dilemmas arise that cannot be solved. For example, there is no reason to believe that, in a different series of experiments and for the same event A , this ratio approaches the same limit. Hence the uniqueness of the probability of the event A is not guaranteed.
3. By this definition, probabilities that are based on our personal belief and knowledge are not justifiable. Thus statements such as the following would be meaningless.
 - The probability that the price of oil will be raised in the next six months is 60%.
 - The probability that the 50,000th decimal figure of the number π is 7 exceeds 10%.
 - The probability that it will snow next Christmas is 30%.
 - The probability that Mozart was poisoned by Salieri is 18%.

In 1900, at the International Congress of Mathematicians in Paris, David Hilbert (1862–1943) proposed 23 problems whose solutions were, in his opinion, crucial to the advancement of mathematics. One of these problems was the axiomatic treatment of the theory of probability. In his lecture, Hilbert quoted Weierstrass, who had said, “The final object, always to be kept

in mind, is to arrive at a correct understanding of the foundations of the science.” Hilbert added that a thorough understanding of special theories of a science is necessary for successful treatment of its foundation. Probability had reached that point and was studied enough to warrant the creation of a firm mathematical foundation. Some work toward this goal had been done by Émile Borel (1871–1956), Serge Bernstein (1880–1968), and Richard von Mises (1883–1953), but it was not until 1933 that Andrei Kolmogorov (1903–1987), a prominent Russian mathematician, successfully axiomatized the theory of probability. In Kolmogorov’s work, which is now universally accepted, three self-evident and indisputable properties of probability (discussed later) are taken as **axioms**, and the entire theory of probability is developed and rigorously based on these axioms. In particular, the existence of a constant p , as the limit of the proportion of the number of times that the event A occurs when the number of experiments increases to ∞ , in some sense, is shown. Subjective probabilities based on our personal knowledge, feelings, and beliefs may also be modeled and studied by this axiomatic approach.

In this book we study the mathematics of probability based on the axiomatic approach. Since in this approach the concepts of **sample space** and **event** play a central role, we now explain these concepts in detail.

1.2 SAMPLE SPACE AND EVENTS

If the outcome of an experiment is not certain but all of its possible outcomes are predictable in advance, then the set of all these possible outcomes is called the **sample space** of the experiment and is usually denoted by S . Therefore, the sample space of an experiment consists of all possible outcomes of the experiment. These outcomes are sometimes called **sample points**, or simply **points**, of the sample space. In the language of probability, certain subsets of S are referred to as **events**. So events are sets of points of the sample space. Some examples follow.

Example 1.1 For the experiment of *tossing a coin once*, the sample space S consists of two points (outcomes), “heads” (H) and “tails” (T). Thus $S = \{H, T\}$. ♦

Example 1.2 Suppose that an experiment consists of two steps. First a coin is flipped. If the outcome is tails, a die is tossed. If the outcome is heads, the coin is flipped again. The sample space of this experiment is $S = \{T1, T2, T3, T4, T5, T6, HT, HH\}$. For this experiment, the event of *heads in the first flip of the coin* is $E = \{HT, HH\}$, and the event of *an odd outcome when the die is tossed* is $F = \{T1, T3, T5\}$. ♦

Example 1.3 Consider measuring the lifetime of a light bulb. Since any nonnegative real number can be considered as the lifetime of the light bulb (in hours), the sample space is $S = \{x: x \geq 0\}$. In this experiment, $E = \{x: x \geq 100\}$ is the event that *the light bulb lasts at least 100 hours*, $F = \{x: x \leq 1000\}$ is the event that *it lasts at most 1000 hours*, and $G = \{505.5\}$ is the event that *it lasts exactly 505.5 hours*. ♦

Example 1.4 Suppose that a study is being done on all families with one, two, or three children. Let the outcomes of the study be the genders of the children in descending order of their ages. Then

$$S = \{b, g, bg, gb, bb, gg, bbb, bgb, bbg, bgg, gbg, ggb, gbb\}.$$

Here the outcome b means that the child is a boy, and g means that it is a girl. The events $F = \{b, bg, bb, bbb, bgb, bbg, bgg\}$ and $G = \{gg, bgg, gbg, ggb\}$ represent families where the eldest child is a boy and families with exactly two girls, respectively. ♦

Example 1.5 A bus with a capacity of 34 passengers stops at a station some time between 11:00 A.M. and 11:40 A.M. every day. The sample space of the experiment, consisting of counting the number of passengers on the bus and measuring the arrival time of the bus, is

$$S = \left\{ (i, t) : 0 \leq i \leq 34, \ 11 \leq t \leq 11\frac{2}{3} \right\}, \quad (1.1)$$

where i represents the number of passengers and t the arrival time of the bus in hours and fractions of hours. The subset of S defined by $F = \{(27, t) : 11\frac{1}{3} < t < 11\frac{2}{3}\}$ is the event that the bus arrives between 11:20 A.M. and 11:40 A.M. with 27 passengers. ♦

Remark 1.1 Different manifestations of outcomes of an experiment might lead to different representations for the sample space of the same experiment. For instance, in [Example 1.5](#), the outcome that the bus arrives at t with i passengers is represented by (i, t) , where t is expressed in hours and fractions of hours. By this representation, (1.1) is the sample space of the experiment. Now if the same outcome is denoted by (i, t) , where t is the number of minutes after 11 A.M. that the bus arrives, then the sample space takes the form

$$S_1 = \{(i, t) : 0 \leq i \leq 34, \ 0 \leq t \leq 40\}.$$

To the outcome that the bus arrives at 11:20 A.M. with 31 passengers, in S the corresponding point is $(31, 11\frac{1}{3})$, while in S_1 it is $(31, 20)$. ♦

Example 1.6 (Round-Off Error) Suppose that each time Jay charges an item to his credit card, he will round the amount to the nearest dollar in his records. Therefore, the round-off error, which is the true value charged minus the amount recorded, is random, with the sample space

$$S = \{0, 0.01, 0.02, \dots, 0.49, -0.50, -0.49, \dots, -0.01\},$$

where we have assumed that for any integer dollar amount a , Jay rounds $a.50$ to $a + 1$. The event of rounding off at most 3 cents in a random charge is given by

$$\{0, 0.01, 0.02, 0.03, -0.01, -0.02, -0.03\}. \quad \blacklozenge$$

If the outcome of an experiment belongs to an event E , we say that the event E has **occurred**. For example, if we draw two cards from an ordinary deck of 52 cards and observe that one is a spade and the other a heart, all of the events $\{sh\}$, $\{sh, dd\}$, $\{cc, dh, sh\}$, $\{hc, sh, ss, hh\}$, and $\{cc, hh, sh, dd\}$ have occurred because sh , the outcome of the experiment, belongs to all of them. However, none of the events $\{dh, sc\}$, $\{dd\}$, $\{ss, hh, cc\}$, and $\{hd, hc, dc, sc, sd\}$ has occurred because sh does not belong to any of them.

In the study of probability theory the relations between different events of an experiment play a central role. In the remainder of this section we study these relations. In all of the following definitions the events belong to a fixed sample space S .

Subset An event E is said to be a **subset** of the event F if, whenever E occurs, F also occurs. This means that all of the sample points of E are contained in F . Hence considering E and F solely as two sets, E is a subset of F in the usual set-theoretic sense: that is, $E \subseteq F$.

Equality Events E and F are said to be **equal** if the occurrence of E implies the occurrence of F , and vice versa; that is, if $E \subseteq F$ and $F \subseteq E$, hence $E = F$.

- Intersection** An event is called the **intersection** of two events E and F if it occurs only whenever E and F occur simultaneously. In the language of sets this event is denoted by EF or $E \cap F$ because it is the set containing exactly the common points of E and F .
- Union** An event is called the **union** of two events E and F if it occurs whenever at least one of them occurs. This event is $E \cup F$ since all of its points are in E or F or both.
- Complement** An event is called the **complement** of the event E if it only occurs whenever E does not occur. The complement of E is denoted by E^c .
- Difference** An event is called the **difference** of two events E and F if it occurs whenever E occurs but F does not. The difference of the events E and F is denoted by $E - F$. It is clear that $E^c = S - E$ and $E - F = EF^c$.
- Certainty** An event is called **certain** if its occurrence is inevitable. Thus the sample space is a certain event.
- Impossibility** An event is called **impossible** if there is certainty in its nonoccurrence. Therefore, the empty set $\emptyset$, which is S^c , is an impossible event.
- Mutually Exclusiveness** If the joint occurrence of two events E and F is impossible, we say that E and F are **mutually exclusive**. Thus E and F are mutually exclusive if the occurrence of E precludes the occurrence of F , and vice versa. Since the event representing the joint occurrence of E and F is EF , their intersection, E and F , are mutually exclusive if $EF = \emptyset$. A set of events $\{E_1, E_2, \dots\}$ is called **mutually exclusive** if the joint occurrence of any two of them is impossible, that is, if $\forall i \neq j, E_i E_j = \emptyset$. Thus $\{E_1, E_2, \dots\}$ is mutually exclusive if and only if every pair of them is mutually exclusive.

The events $\bigcup_{i=1}^n E_i$, $\bigcap_{i=1}^n E_i$, $\bigcup_{i=1}^{\infty} E_i$, and $\bigcap_{i=1}^{\infty} E_i$ are defined in a way similar to $E_1 \cup E_2$ and $E_1 \cap E_2$. For example, if $\{E_1, E_2, \dots, E_n\}$ is a set of events, by $\bigcup_{i=1}^n E_i$ we mean the event in which at least one of the events E_i , $1 \leq i \leq n$, occurs. By $\bigcap_{i=1}^n E_i$ we mean an event that occurs only when all of the events E_i , $1 \leq i \leq n$, occur.

Example 1.7 At a busy international airport, arriving planes land on a first-come, first-served basis. Let

E = there are at least five planes waiting to land,

F = there are at most three planes waiting to land,

H = there are exactly two planes waiting to land.

Then

1. E^c is the event that at most four planes are waiting to land.
2. F^c is the event that at least four planes are waiting to land.
3. E is a subset of F^c ; that is, if E occurs, then F^c occurs. Therefore, $EF^c = E$.
4. H is a subset of F ; that is, if H occurs, then F occurs. Therefore, $FH = H$.
5. E and F are mutually exclusive; that is, $EF = \emptyset$. E and H are also mutually exclusive since $EH = \emptyset$.

6. FH^c is the event that the number of planes waiting to land is zero, one, or three. ♦

Sometimes Venn diagrams are used to represent the relations among events of a sample space. The sample space S of the experiment is usually shown as a large rectangle and, inside S , circles or other geometrical objects are drawn to indicate the events of interest. Figure 1.1 presents Venn diagrams for EF , $E \cup F$, E^c , and $(E^c G) \cup F$. The shaded regions are the indicated events.

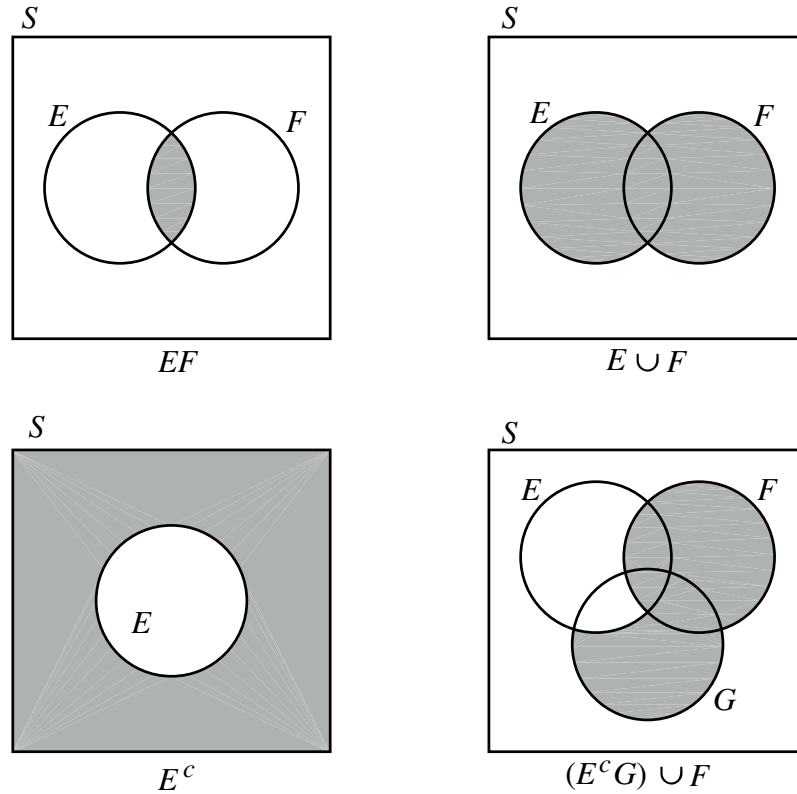


Figure 1.1 Venn diagrams of the events specified.

Example 1.8 Figure 1.2 shows an electric circuit in which, at any random time, each of the switches located at 1, 2, 3, 4, and 5 is either closed or open. For $1 \leq i \leq 5$, let E_i be the event that, at a random time, the switch at location i is closed. In terms of E_i 's, describe the event that a signal fed to the input is transmitted to the output.

Solution: The event that a signal fed to the input is transmitted to the output is

$$F = E_1 E_5 \cup E_1 E_3 E_4 \cup E_2 E_4 \cup E_2 E_3 E_5. \quad \blacklozenge$$

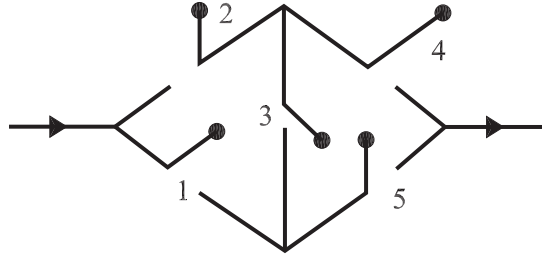


Figure 1.2 Electric circuit of [Example 1.8](#).

Unions, intersections, and complementations satisfy many useful relations between events. A few of these relations are as follows:

$$(E^c)^c = E, \quad E \cup E^c = S, \quad ES = E, \quad \text{and} \quad EE^c = \emptyset.$$

Commutative laws:

$$E \cup F = F \cup E, \quad EF = FE.$$

Associative laws:

$$E \cup (F \cup G) = (E \cup F) \cup G, \quad E(FG) = (EF)G.$$

Distributive laws:

$$(EF) \cup H = (E \cup H)(F \cup H), \quad (E \cup F)H = (EH) \cup (FH).$$

De Morgan's first law:

$$(E \cup F)^c = E^c F^c, \quad \left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c, \quad \left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c.$$

De Morgan's second law:

$$(EF)^c = E^c \cup F^c, \quad \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c, \quad \left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c.$$

Another useful relation between E and F , two arbitrary events of a sample space S , is

$$E = EF \cup EF^c.$$

This equality readily follows from $E = ES$ and distributivity:

$$E = ES = E(F \cup F^c) = EF \cup EF^c.$$

These and similar identities are usually proved by the **elementwise method**. The idea is to show that the events on both sides of the equation are formed of the same sample points. To use this method, we prove set inclusion in both directions. That is, sample points belonging to the event on the left also belong to the event on the right, and vice versa. An example follows.

Example 1.9 Prove De Morgan's first law: For E and F , two events of a sample space S , $(E \cup F)^c = E^c F^c$.

Proof: First we show that $(E \cup F)^c \subseteq E^c F^c$; then we prove the reverse inclusion $E^c F^c \subseteq (E \cup F)^c$. To show that $(E \cup F)^c \subseteq E^c F^c$, let x be an outcome that belongs to $(E \cup F)^c$. Then x does not belong to $E \cup F$, meaning that x is neither in E nor in F . So x belongs to both E^c and F^c and hence to $E^c F^c$. To prove the reverse inclusion, let $x \in E^c F^c$. Then $x \in E^c$ and $x \in F^c$, implying that $x \notin E$ and $x \notin F$. Therefore, $x \notin E \cup F$ and thus $x \in (E \cup F)^c$. ♦

Note that Venn diagrams are an excellent way to give intuitive justification for the validity of relations or to create counterexamples and show invalidity of relations. However, they are not appropriate to prove relations. This is because of the large number of cases that must be considered (particularly if more than two events are involved). For example, suppose that by means of Venn diagrams, we want to prove the identity $(EF)^c = E^c \cup F^c$. First we must draw appropriate representations for all possible ways that E and F can be related: cases such as $EF = \emptyset$, $EF \neq \emptyset$, $E = F$, $E = \emptyset$, $F = S$, and so on. Then in each particular case we should find the regions that represent $(EF)^c$ and $E^c \cup F^c$ and observe that they are the same. Even if these two sets have different representations in only one case, the identity would be false.

EXERCISES

A

1. From the letters of the word MISSISSIPPI, a letter is chosen at random. What is a sample space for this experiment? What is the event that the outcome is a vowel?
2. Last month, an insurance company sold 57 life insurance policies. Define a sample space for the number of claims that the company will receive from the beneficiaries of this group within the next 15 years. What is the event that the company receives at least 3 but no more than 8 claims?
3. In the experiment of flipping a coin three times, what does the event

$$E = \{\text{THH, HTH, HHT, HHH}\}$$

represent?

4. In the experiment of tossing two dice, what do the following events represent?
 $E = \{(1, 3), (2, 6), (3, 1), (6, 2)\}$ and $F = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$.
5. A deck of six cards consists of three black cards numbered 1, 2, 3, and three red cards numbered 1, 2, 3. First, Vann draws a card at random and without replacement. Then Paul draws a card at random and without replacement from the remaining cards. Let A be the event that Paul's card has a larger number than Vann's card. Let B be the event that Vann's card has a larger number than Paul's card.

- (a) Are A and B mutually exclusive?

- (b) Are A and B complements of one another?
6. A box contains three red and five blue balls. Define a sample space for the experiment of recording the colors of three balls that are drawn from the box, one by one, with replacement.
 7. Define a sample space for the experiment of choosing a number from the interval $(0, 20)$. Describe the event that such a number is an integer.
 8. Define a sample space for the experiment of putting three different books on a shelf in random order. If two of these three books are a two-volume dictionary, describe the event that these volumes stand in increasing order side-by-side (i.e., volume I precedes volume II).
 9. Two dice are rolled. Let E be the event that the sum of the outcomes is odd and F be the event of at least one 1. Interpret the events EF , E^cF , and E^cF^c .
 10. Define a sample space for the experiment of drawing two coins from a purse that contains two quarters, three nickels, one dime, and four pennies. For the same experiment describe the following events:
 - (a) drawing 26 cents;
 - (b) drawing more than 9 but less than 25 cents;
 - (c) drawing 29 cents.
 11. A telephone call from a certain person is received some time between 7:00 A.M. and 9:10 A.M. every day. Define a sample space for this phenomenon, and describe the event that the call arrives within 15 minutes of the hour.
 12. A device that has two components fails if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 \leq i \leq 2$, denote the outcome that the i th component is operative at the random time. In terms of A_i 's, (a) define a sample space for the status of the system at the random time; (b) describe the event that the device is not operative at that random time.
 13. When flipping a coin more than once, what experiment has a sample space defined by the following?

$$S = \{TT, HTT, THTT, HHTT, HHHTT, HTHTT, THHTT, \dots\}.$$
 14. Let E , F , and G be three events; explain the meaning of the relations $E \cup F \cup G = G$ and $EF G = G$.
 15. A limousine that carries passengers from an airport to three different hotels just left the airport with two passengers. Describe the sample space of the stops and the event that both of the passengers get off at the same hotel.
 16. A psychologist, who is interested in human complexion, in a study of hues and shades, shows her subjects three pieces of wood colored almond, lemon, and flax, respectively. She then asks them to identify their favorite colors. Define a sample space for the answer given by a random subject.
 17. For a saw blade manufacturer's products, the global demand, per month, for band saws is between 30 and 36 thousands; for reciprocating saws, it is between 28 and 33 thousands;

and for hole saws, it is between 300 and 600 thousands. Define a sample space for the demands for these three types of saws by this manufacturer in a random month. Describe the event that, in such a month, the demand for band saws is at least 33 thousands and for hole saws, it is less than 435 thousands. In this exercise, by “between two numbers a and b ,” we mean inclusive of both a and b .

18. An insurance company sells a joint life insurance policy to Alexia and her husband, Roy. Define a sample space for the death or survival of this couple in five years. What is the event that at that time only one of them lives?
19. Find the simplest possible expression for the following events.
 - (a) $(E \cup F)(F \cup G)$.
 - (b) $(E \cup F)(E^c \cup F)(E \cup F^c)$.
20. At a certain university, every year eight to 12 professors are granted University Merit Awards. This year among the nominated faculty are Drs. Jones, Smith, and Brown. Let A , B , and C denote the events, respectively, that these professors will be given awards. In terms of A , B , and C , find an expression for the event that the award goes to (a) only Dr. Jones; (b) at least one of the three; (c) none of the three; (d) exactly two of them; (e) exactly one of them; (f) Drs. Jones or Smith but not both.
21. A device that has n , $n > 1$, components fails to operate if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 < i \leq n$, denote the event that the i th component is operative at the random time. In terms of A_i 's, describe the event that the device is operative at that random time.
22. Prove that the event B is impossible if and only if for every event A ,

$$A = (B \cap A^c) \cup (B^c \cap A).$$

23. Let E , F , and G be three events. Determine which of the following statements are correct and which are incorrect. Justify your answers.
 - (a) $(E - EF) \cup F = E \cup F$.
 - (b) $F^c G \cup E^c G = G(F \cup E)^c$.
 - (c) $(E \cup F)^c G = E^c F^c G$.
 - (d) $EF \cup EG \cup FG \subset E \cup F \cup G$.
24. Travis picks up darts to shoot toward an 18"-diameter dartboard aiming at the bullseye on the board. Describe a sample space for the point at which a dart hits the board.
25. For the experiment of flipping a coin until a heads occurs, (a) describe the sample space; (b) describe the event that it takes an odd number of flips until a heads occurs.
26. In an experiment, cards are drawn, one by one, at random and successively from an ordinary deck of 52 cards. Let A_n be the event that no face card or ace appears on the first $n - 1$ drawings, and the n th draw is an ace. In terms of A_n 's, find an expression for the event that an ace appears before a face card, (a) if the cards are drawn with replacement; (b) if they are drawn without replacement.
27. A point is chosen at random from the interval $(-1, 1)$. Let E_1 be the event that it falls in the interval $(-1/2, 1/2)$, E_2 be the event that it falls in the interval $(-1/4, 1/4)$ and, in

general, for $1 \leq i < \infty$, E_i be the event that the point is in the interval $(-1/2^i, 1/2^i)$. Find $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$.

B

28. Prove De Morgan's second law, $(AB)^c = A^c \cup B^c$, (a) by elementwise proof; (b) by applying De Morgan's first law to A^c and B^c .
29. Let A and B be two events. Prove the following relations by the elementwise method.
 - (a) $(A - AB) \cup B = A \cup B$.
 - (b) $(A \cup B) - AB = AB^c \cup A^c B$.
30. Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of events. Prove that for every event B ,
 - (a) $B(\bigcup_{i=1}^{\infty} A_i) = \bigcup_{i=1}^{\infty} BA_i$.
 - (b) $B \cup (\bigcap_{i=1}^{\infty} A_i) = \bigcap_{i=1}^{\infty} (B \cup A_i)$.
31. Define a sample space for the experiment of putting in a random order seven different books on a shelf. If three of these seven books are a three-volume dictionary, describe the event that these volumes stand in increasing order side by side (i.e., volume I precedes volume II and volume II precedes volume III).
32. In a mathematics department of 31 voting faculty members, there are three candidates running for the chair position. The voting procedure adopted by the department is **approval voting**, in which the eligible voters can vote for as many candidates as they wish. The candidate with the maximum votes will win. Let A_{ij} , $1 \leq i \leq 31$, $1 \leq j \leq 3$, be the event that, in the next election, the i th faculty member votes for the j th candidate. In terms of A_{ij} 's, describe the sample space for the next election of this department.
33. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events. Find an expression for the event that infinitely many of the A_i 's occur.
34. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events of a sample space S . Find a sequence $\{B_1, B_2, B_3, \dots\}$ of mutually exclusive events such that for all $n \geq 1$, $\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n B_i$.



Self-Quiz on Section 1.2

Time allotted: 20 Minutes

Each problem is worth 2.5 points.

1. Jody, Ann, Bill, and Karl line up in a random order to get their photo taken. Describe the event that, on the line, males and females alternate.
2. In a large hospital, there are 100 patients scheduled to have heart bypass surgery. Let E_i , $1 \leq i \leq 100$, be the event that the i th patient lives through the postoperative period of the surgery. In terms of E_i 's, (a) describe the event that all patients survive the critical postoperative period; (b) describe the event that exactly one patient dies during that period.

3. Find the simplest possible expression for the event $(E \cup F)(F \cup G)(EG \cup F^c)$.
4. Consider the system shown by the diagram of Figure 1.3, consisting of 7 components denoted by 1, 2, ..., 7. Suppose that each component is either functioning or not functioning, with no other capabilities. Suppose that the system itself also has two performance capabilities, functioning and not functioning. For $1 \leq i \leq 7$, let E_i be the event that component i is functioning. In terms of E_i 's, describe the event that a signal fed to the input is transmitted to the output.

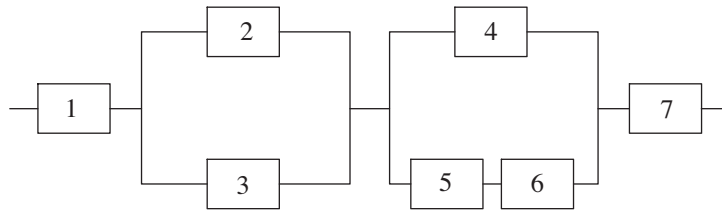


Figure 1.3 A diagram for the system of Self-Quiz Problem 4.

1.3 AXIOMS OF PROBABILITY

In mathematics, the goals of researchers are to obtain new results and prove their correctness, create simple proofs for already established results, discover or create connections between different fields of mathematics, construct and solve mathematical models for real-world problems, and so on. To discover new results, mathematicians use trial and error, instinct and inspired guessing, inductive analysis, studies of special cases, and other methods. But when a new result is discovered, its validity remains subject to skepticism until it is rigorously proven. Sometimes attempts to prove a result fail and contradictory examples are found. Such examples that invalidate a result are called **counterexamples**. No mathematical proposition is settled unless it is either proven or refuted by a counterexample. If a result is false, a counterexample exists to refute it. Similarly, if a result is valid, a proof must be found for its validity, although in some cases it might take years, decades, or even centuries to find it.

Proofs in probability theory (and virtually any other theory) are done in the framework of the **axiomatic method**. By this method, if we want to convince any rational person, say Sonya, that a statement L_1 is correct, we will show her how L_1 can be deduced logically from another statement L_2 that might be acceptable to her. However, if Sonya does not accept L_2 , we should demonstrate how L_2 can be deduced logically from a simpler statement L_3 . If she disputes L_3 , we must continue this process until, somewhere along the way we reach a statement that, without further justification, is acceptable to her. This statement will then become the basis of our argument. Its existence is necessary since otherwise the process continues ad infinitum without any conclusions. Therefore, in the axiomatic method, first we adopt certain simple, indisputable, and consistent statements without justifications. These are **axioms** or **postulates**. Then we agree on how and when one statement is a logical consequence of another one and, finally, using the terms that are already clearly understood, axioms and definitions, we obtain new results. New results found in this manner are called **theorems**. Theorems are statements

that can be proved. Upon establishment, they are used for discovery of new theorems, and the process continues and a theory evolves.

In this book, our approach is based on the axiomatic method. There are three axioms upon which probability theory is based and, except for them, everything else needs to be proved. We will now explain these axioms.

Definition 1.1 (Probability Axioms) *Let S be the sample space of a random phenomenon. Suppose that to each event A of S , a number denoted by $P(A)$ is associated with A . If P satisfies the following axioms, then it is called a **probability** and the number $P(A)$ is said to be the **probability of A** .*

Axiom 1 $P(A) \geq 0$.

Axiom 2 $P(S) = 1$.

Axiom 3 *If $\{A_1, A_2, A_3, \dots\}$ is a sequence of mutually exclusive events (i.e., the joint occurrence of every pair of them is impossible: $A_i A_j = \emptyset$ when $i \neq j$), then*

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Note that the axioms of probability are a set of rules that must be satisfied before S and P can be considered a probability model.

Axiom 1 states that the probability of the occurrence of an event is always nonnegative. Axiom 2 guarantees that the probability of the occurrence of the event S that is certain is 1. Axiom 3 states that for a sequence of mutually exclusive events the probability of the occurrence of at least one of them is equal to the sum of their probabilities.

Axiom 2 is merely a convenience to make things definite. It would be equally reasonable to have $P(S) = 100$ and interpret probabilities as percentages (which we frequently do).

Let S be the sample space of an experiment. Let A and B be events of S . We say that A and B are **equally likely** if $P(A) = P(B)$. Let ω_1 and ω_2 be sample points of S . We say that ω_1 and ω_2 are **equally likely** if the events $\{\omega_1\}$ and $\{\omega_2\}$ are equally likely, that is, if $P(\{\omega_1\}) = P(\{\omega_2\})$.

We will now prove some immediate implications of the axioms of probability.

Theorem 1.1 *The probability of the empty set $\emptyset$ is 0. That is, $P(\emptyset) = 0$.*

Proof: Let $A_1 = S$ and $A_i = \emptyset$ for $i \geq 2$; then $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events. Thus, by Axiom 3,

$$P(S) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = P(S) + \sum_{i=2}^{\infty} P(\emptyset),$$

implying that $\sum_{i=2}^{\infty} P(\emptyset) = 0$. This is possible only if $P(\emptyset) = 0$. ♦

Axiom 3 is stated for a *countably infinite* collection of mutually exclusive events. For this reason, it is also called the *axiom of countable additivity*. We will now show that the same property holds for a *finite* collection of mutually exclusive events as well. That is, P also satisfies *finite additivity*.

Theorem 1.2 *Let $\{A_1, A_2, \dots, A_n\}$ be a mutually exclusive set of events. Then*

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

Proof: For $i > n$, let $A_i = \emptyset$. Then $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events. From Axiom 3 and [Theorem 1.1](#) we get

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \\ &= \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(A_i) = \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(\emptyset) \\ &= \sum_{i=1}^n P(A_i). \quad \blacklozenge \end{aligned}$$

Letting $n = 2$, [Theorem 1.2](#) implies that if A_1 and A_2 are mutually exclusive, then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2). \quad (1.2)$$

The intuitive meaning of (1.2) is that if an experiment can be repeated indefinitely, then for two mutually exclusive events A_1 and A_2 , the proportion of times that $A_1 \cup A_2$ occurs is equal to the sum of the proportion of times that A_1 occurs and the proportion of times that A_2 occurs. For example, for the experiment of tossing a fair die, $S = \{1, 2, 3, 4, 5, 6\}$ is the sample space. Let A_1 be the event that the outcome is 6, and A_2 be the event that the outcome is odd. Then $A_1 = \{6\}$ and $A_2 = \{1, 3, 5\}$. Since all sample points are equally likely to occur (by the definition of a fair die) and the number of sample points of A_1 is $1/6$ of the number of sample points of S , we expect that $P(A_1) = 1/6$. Similarly, we expect that $P(A_2) = 3/6$. Now $A_1 A_2 = \emptyset$ implies that the number of sample points of $A_1 \cup A_2$ is $(1/6 + 3/6)$ th of the number of sample points of S . Hence we should expect that $P(A_1 \cup A_2) = 1/6 + 3/6$, which is the same as $P(A_1) + P(A_2)$. This and many other examples suggest that (1.2) is a reasonable relation to be taken as Axiom 3. However, if we do this, difficulties arise when a sample space contains infinitely many sample points, that is, when the number of possible outcomes of an experiment is not finite. For example, in successive throws of a die let A_n be the event that the first 6 occurs on the n th throw. Then we would be unable to find the probability of $\bigcup_{n=1}^{\infty} A_n$, which represents the event that eventually a 6 occurs. For this reason, Axiom 3, which is the infinite analog of (1.2), is required. It by no means contradicts our intuitive ideas of probability, and one of its great advantages is that [Theorems 1.1](#) and [1.2](#) are its immediate consequences.

A significant implication of (1.2) is that for any event A , $P(A) \leq 1$. To see this, note that

$$P(A \cup A^c) = P(A) + P(A^c).$$

Now, by Axiom 2,

$$P(A \cup A^c) = P(S) = 1;$$

therefore, $P(A) + P(A^c) = 1$. This and Axiom 1 imply that $P(A) \leq 1$. Hence

The probability of the occurrence of an event is always some number between 0 and 1. That is,

$$0 \leq P(A) \leq 1.$$

★ **Remark 1.2[†]** Let S be the sample space of an experiment. The set of all subsets of S is denoted by $\mathcal{P}(S)$ and is called the **power set** of S . Since the aim of probability theory is to associate a number between 0 and 1 to every subset of the sample space, probability is a function P from $\mathcal{P}(S)$ to $[0, 1]$. However, in theory, there is one exception to this: If the sample space S is not countable, not all of the elements of $\mathcal{P}(S)$ are events. There are elements of $\mathcal{P}(S)$ that are not (in a sense defined in more advanced probability texts) **measurable**. These elements are not events (see [Example 1.22](#)). In other words, it is a curious mathematical fact that the Kolmogorov axioms are inconsistent with the notion that every subset of every sample space has a probability. Since in real-world problems we are only dealing with those elements of $\mathcal{P}(S)$ that are measurable, we are not concerned with these exceptions. We must also add that, in general, if the domain of a function is a collection of sets, it is called a **set function**. Hence probability is a real-valued, nonnegative, countably additive set function. ♦

Example 1.10 A coin is called **unbiased** or **fair** if, whenever it is flipped, the probability of obtaining heads equals that of obtaining tails. Suppose that in an experiment an unbiased coin is flipped. The sample space of such an experiment is $S = \{T, H\}$. Since the events $\{H\}$ and $\{T\}$ are equally likely to occur, $P(\{T\}) = P(\{H\})$, and since they are mutually exclusive,

$$P(\{T, H\}) = P(\{T\}) + P(\{H\}).$$

Hence Axioms 2 and 3 imply that

$$1 = P(S) = P(\{H, T\}) = P(\{H\}) + P(\{T\}) = P(\{H\}) + P(\{H\}) = 2P(\{H\}).$$

This gives that $P(\{H\}) = 1/2$ and $P(\{T\}) = 1/2$. Now suppose that an experiment consists of flipping a biased coin where the outcome of tails is twice as likely as heads; then $P(\{T\}) = 2P(\{H\})$. Hence

$$1 = P(S) = P(\{H, T\}) = P(\{H\}) + P(\{T\}) = P(\{H\}) + 2P(\{H\}) = 3P(\{H\}).$$

This shows that $P(\{H\}) = 1/3$; thus $P(\{T\}) = 2/3$. ♦

Example 1.11 Sharon has baked five loaves of bread that are identical except that one of them is underweight. Sharon's husband chooses one of these loaves at random. Let B_i , $1 \leq i \leq 5$, be the event that he chooses the i th loaf. Since all five loaves are equally likely to be drawn, we have

$$P(\{B_1\}) = P(\{B_2\}) = P(\{B_3\}) = P(\{B_4\}) = P(\{B_5\}).$$

But the events $\{B_1\}$, $\{B_2\}$, $\{B_3\}$, $\{B_4\}$, and $\{B_5\}$ are mutually exclusive, and the sample space is $S = \{B_1, B_2, B_3, B_4, B_5\}$. Therefore, by Axioms 2 and 3,

$$1 = P(S) = P(\{B_1\}) + P(\{B_2\}) + P(\{B_3\}) + P(\{B_4\}) + P(\{B_5\}) = 5 \cdot P(\{B_1\}).$$

This gives $P(\{B_1\}) = 1/5$ and hence $P(\{B_i\}) = 1/5$, $1 \leq i \leq 5$. Therefore, the probability that Sharon's husband chooses the underweight loaf is $1/5$. ♦

From [Examples 1.10](#) and [1.11](#) it should be clear that if a sample space contains N points that are equally likely to occur, then the probability of each outcome (sample point) is $1/N$.

[†]Throughout the book, items that are optional and can be skipped are identified by ★'s.

In general, this can be shown as follows. Let $S = \{s_1, s_2, \dots, s_N\}$ be the sample space of an experiment; then, if all of the sample points are equally likely to occur, we have

$$P(\{s_1\}) = P(\{s_2\}) = \dots = P(\{s_N\}).$$

But $P(S) = 1$, and the events $\{s_1\}, \{s_2\}, \dots, \{s_N\}$ are mutually exclusive. Therefore,

$$\begin{aligned} 1 = P(S) &= P(\{s_1, s_2, \dots, s_N\}) \\ &= P(\{s_1\}) + P(\{s_2\}) + \dots + P(\{s_N\}) = NP(\{s_1\}). \end{aligned}$$

This shows that $P(\{s_1\}) = 1/N$. Thus $P(\{s_i\}) = 1/N$ for $1 \leq i \leq N$.

One simple consequence of the axioms of probability is that, if the sample space of an experiment contains N points that are all equally likely to occur, then the probability of the occurrence of any event A is equal to the number of points of A , say $N(A)$, divided by N . Historically, until the introduction of the axiomatic method by A. N. Kolmogorov in 1933, this fact was taken as the definition of the probability of A . It is now called the **classical definition of probability**. The following theorem, which shows that the classical definition is a simple result of the axiomatic approach, is also an important tool for the computation of probabilities of events for experiments with finite sample spaces.

Theorem 1.3 *Let S be the sample space of an experiment. If S has N points that are all equally likely to occur, then for any event A of S ,*

$$P(A) = \frac{N(A)}{N},$$

where $N(A)$ is the number of points of A .

Proof: Let $S = \{s_1, s_2, \dots, s_N\}$, where each s_i is an outcome (a sample point) of the experiment. Since the outcomes are equiprobable, $P(\{s_i\}) = 1/N$ for all i , $1 \leq i \leq N$. Now let $A = \{s_{i_1}, s_{i_2}, \dots, s_{i_{N(A)}}\}$, where $s_{i_j} \in S$ for all i_j . Since $\{s_{i_1}\}, \{s_{i_2}\}, \dots, \{s_{i_{N(A)}}\}$ are mutually exclusive, Axiom 3 implies that

$$\begin{aligned} P(A) &= P(\{s_{i_1}, s_{i_2}, \dots, s_{i_{N(A)}}\}) \\ &= P(\{s_{i_1}\}) + P(\{s_{i_2}\}) + \dots + P(\{s_{i_{N(A)}}\}) \\ &= \underbrace{\frac{1}{N} + \frac{1}{N} + \dots + \frac{1}{N}}_{N(A) \text{ terms}} = \frac{N(A)}{N}. \quad \blacklozenge \end{aligned}$$

Example 1.12 Let S be the sample space of flipping a fair coin three times and A be the event of at least two heads; then

$$S = \{HHH, HTH, HHT, HTT, THH, THT, TTH, TTT\}$$

and $A = \{HHH, HTH, HHT, THH\}$. So $N = 8$ and $N(A) = 4$. Therefore, the probability of at least two heads in flipping a fair coin three times is $N(A)/N = 4/8 = 1/2$. $\blacklozenge$

Example 1.13 An elevator with two passengers stops at the second, third, and fourth floors. If it is equally likely that a passenger gets off at any of the three floors, what is the probability that the passengers get off at different floors?

Solution: Let a and b denote the two passengers and a_2b_4 mean that a gets off at the second floor and b gets off at the fourth floor, with similar representations for other cases. Let A be the event that the passengers get off at different floors. Then

$$S = \{a_2b_2, a_2b_3, a_2b_4, a_3b_2, a_3b_3, a_3b_4, a_4b_2, a_4b_3, a_4b_4\}$$

and $A = \{a_2b_3, a_2b_4, a_3b_2, a_3b_4, a_4b_2, a_4b_3\}$. So $N = 9$ and $N(A) = 6$. Therefore, the desired probability is $N(A)/N = 6/9 = 2/3$. ♦

Example 1.14 A number is selected at random from the set of integers $\{1, 2, \dots, 1000\}$. What is the probability that the number is divisible by 3?

Solution: Here the sample space contains 1000 points, so $N = 1000$. Let A be the set of all numbers between 1 and 1000 that are divisible by 3. Then $A = \{3m : 1 \leq m \leq 333\}$. So $N(A) = 333$. Therefore, the probability that a random natural number between 1 and 1000 is divisible by 3 is equal to $333/1000$. ♦

Example 1.15 A number is selected at random from the set $\{1, 2, \dots, N\}$. What is the probability that the number is divisible by k , $1 \leq k \leq N$?

Solution: Here the sample space contains N points. Let A be the event that the outcome is divisible by k . Then $A = \{km : 1 \leq m \leq [N/k]\}$, where $[N/k]$ is the greatest integer less than or equal to N/k (to compute $[N/k]$, just divide N by k and round down). So $N(A) = [N/k]$ and $P(A) = [N/k]/N$. ♦

Remark 1.3 As explained in [Remark 1.1](#), different manifestations of outcomes of an experiment may lead to different representations for the sample space of the same experiment. Because of this, different sample points of a representation might not have the same probability of occurrence. For example, suppose that a study is being done on families with three children. Let the outcomes of the study be the number of girls and the number of boys in a randomly selected family. Then $S = \{bbb, bbg, bgb, bbb, ggb, gbg, gbb, ggg\}$ and $\Omega = \{bbb, bbg, bgg, ggg\}$ are both reasonable sample spaces for the genders of the children of the family. In S , for example, bgg means that the first child of the family is a boy, the second child is a girl, and the third child is also a girl. In Ω , bgg means that the family has one boy and two girls. Therefore, in S all sample points occur with the same probability, namely, $1/8$. In Ω , however, probabilities associated to the sample points are not equal: $P(\{bbb\}) = P(\{ggg\}) = 1/8$ whereas $P(\{bbg\}) = P(\{bgg\}) = 3/8$. ♦

Finally, we should note that for finite sample spaces, if nonnegative probabilities are assigned to sample points so that they sum to 1, then the probability axioms hold. Let

$$S = \{w_1, w_2, \dots, w_n\}$$

be a sample space. Let $p_1, p_2, \dots, p_n$ be n nonnegative real numbers with $\sum_{i=1}^n p_i = 1$. Let P be defined on subsets of S by $P(\{w_i\}) = p_i$, and

$$P(\{w_{i_1}, w_{i_2}, \dots, w_{i_\ell}\}) = p_{i_1} + p_{i_2} + \dots + p_{i_\ell}.$$

It is straightforward to verify that P satisfies the probability axioms. Hence P defines a probability on the sample space S .

1.4 BASIC THEOREMS

Theorem 1.4 For any event A , $P(A^c) = 1 - P(A)$.

Proof: Since $AA^c = \emptyset$, A and A^c are mutually exclusive. Thus

$$P(A \cup A^c) = P(A) + P(A^c).$$

But $A \cup A^c = S$ and $P(S) = 1$, so

$$1 = P(S) = P(A \cup A^c) = P(A) + P(A^c).$$

Therefore, $P(A^c) = 1 - P(A)$. ♦

This theorem states that the probability of nonoccurrence of the event A is 1 minus the probability of its occurrence. For example, consider $S = \{(i, j): 1 \leq i \leq 6, 1 \leq j \leq 6\}$, the sample space of tossing two fair dice. If A is the event of getting a sum of 4, then $A = \{(1, 3), (2, 2), (3, 1)\}$ and $P(A) = 3/36$. Theorem 1.4 states that the probability of A^c , the event of not getting a sum of 4, which is harder to count, is $1 - 3/36 = 33/36$. As another example, consider the experiment of selecting a random number from the set $\{1, 2, 3, \dots, 1000\}$. By Example 1.14, the probability that the number selected is divisible by 3 is $333/1000$. Thus by Theorem 1.4, the probability that it is not divisible by 3, a quantity harder to find directly, is $1 - 333/1000 = 667/1000$.

Theorem 1.5 If $A \subseteq B$, then

$$P(B - A) = P(BA^c) = P(B) - P(A).$$

Proof: $A \subseteq B$ implies that $B = (B - A) \cup A$ (see Figure 1.4). But $(B - A)A = \emptyset$. So the events $B - A$ and A are mutually exclusive, and $P(B) = P((B - A) \cup A) = P(B - A) + P(A)$. This gives $P(B - A) = P(B) - P(A)$. ♦

Corollary If $A \subseteq B$, then $P(A) \leq P(B)$.

Proof: By Theorem 1.5, $P(B - A) = P(B) - P(A)$. Since $P(B - A) \geq 0$, we have that $P(B) - P(A) \geq 0$. Hence $P(B) \geq P(A)$. ♦

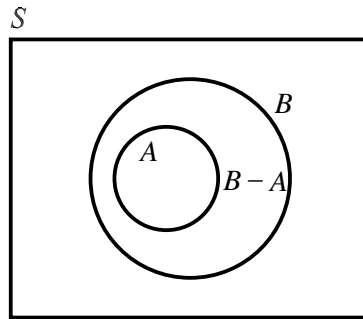


Figure 1.4 $A \subseteq B$ implies that $B = (B - A) \cup A$.

This corollary says that, for instance, it is less likely that a computer has one defect than it has at least one defect. Note that in [Theorem 1.5](#), the condition of $A \subseteq B$ is necessary. The relation $P(B - A) = P(B) - P(A)$ is not true in general. For example, in rolling a fair die, let $B = \{1, 2\}$ and $A = \{3, 4, 5\}$, then $B - A = \{1, 2\}$. Therefore, $P(B - A) = 1/3$, $P(B) = 1/3$, and $P(A) = 1/2$. Hence $P(B - A) \neq P(B) - P(A)$.

Theorem 1.6 $P(A \cup B) = P(A) + P(B) - P(AB)$.

Proof: $A \cup B = A \cup (B - AB)$ (see [Figure 1.5](#)) and $A(B - AB) = \emptyset$, so A and $B - AB$ are mutually exclusive events and

$$P(A \cup B) = P(A \cup (B - AB)) = P(A) + P(B - AB). \quad (1.3)$$

Now since $AB \subseteq B$, [Theorem 1.5](#) implies that

$$P(B - AB) = P(B) - P(AB).$$

Therefore, (1.3) gives

$$P(A \cup B) = P(A) + P(B) - P(AB). \quad \blacklozenge$$

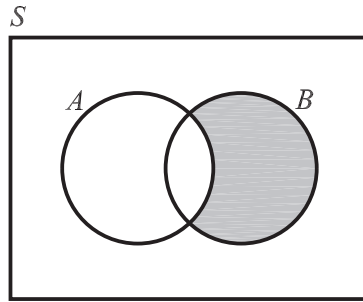


Figure 1.5 The shaded region is $B - AB$.
Thus $A \cup B = A \cup (B - AB)$.

Example 1.16 Suppose that in a community of 400 adults, 300 bike or swim or do both, 160 swim, and 120 swim and bike. What is the probability that an adult, selected at random from this community, bikes?

Solution: Let A be the event that the person swims and B be the event that he or she bikes; then $P(A \cup B) = 300/400$, $P(A) = 160/400$, and $P(AB) = 120/400$. Hence the relation $P(A \cup B) = P(A) + P(B) - P(AB)$ implies that

$$\begin{aligned} P(B) &= P(A \cup B) + P(AB) - P(A) \\ &= \frac{300}{400} + \frac{120}{400} - \frac{160}{400} = \frac{260}{400} = 0.65. \quad \blacklozenge \end{aligned}$$

Example 1.17 A number is chosen at random from the set of integers $\{1, 2, \dots, 1000\}$. What is the probability that it is divisible by 3 or 5 (i.e., either 3 or 5 or both)?

Solution: The number of integers between 1 and N that are divisible by k is computed by dividing N by k and then rounding down (see [Examples 1.15](#)). Therefore, if A is the event that the outcome is divisible by 3 and B is the event that it is divisible by 5, then $P(A) = 333/1000$ and $P(B) = 200/1000$. Now AB is the event that the outcome is divisible by both 3 and 5. Since a number is divisible by 3 and 5 if and only if it is divisible by 15 (3 and 5 are prime numbers), $P(AB) = 66/1000$ (divide 1000 by 15 and round down to get 66). Thus the desired probability is computed as follows:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(AB) \\ &= \frac{333}{1000} + \frac{200}{1000} - \frac{66}{1000} = \frac{467}{1000} = 0.467. \quad \blacklozenge \end{aligned}$$

Theorem 1.6 gives a formula to calculate the probability that at least one of A and B occurs. We may also calculate the probability that at least one of the events $A_1, A_2, A_3, \dots$, and A_n occurs. For three events A_1, A_2 , and A_3 ,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) - P(A_1 A_2) \\ &\quad - P(A_1 A_3) - P(A_2 A_3) + P(A_1 A_2 A_3). \end{aligned}$$

For four events,

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3 \cup A_4) &= P(A_1) + P(A_2) + P(A_3) + P(A_4) - P(A_1 A_2) \\ &\quad - P(A_1 A_3) - P(A_1 A_4) - P(A_2 A_3) - P(A_2 A_4) \\ &\quad - P(A_3 A_4) + P(A_1 A_2 A_3) + P(A_1 A_2 A_4) \\ &\quad + P(A_1 A_3 A_4) + P(A_2 A_3 A_4) - P(A_1 A_2 A_3 A_4). \end{aligned}$$

We now explain a procedure, which will enable us to find $P(A_1 \cup A_2 \cup \dots \cup A_n)$, the probability that at least one of the events $A_1, A_2, \dots, A_n$ occurs.

Inclusion-Exclusion Principle *To calculate $P(A_1 \cup A_2 \cup \dots \cup A_n)$, first find all of the possible intersections of events from $A_1, A_2, \dots, A_n$ and calculate their probabilities. Then add the probabilities of those intersections that are formed of an odd number of events, and subtract the probabilities of those formed of an even number of events.*

The following formula is an expression for the principle of inclusion-exclusion. It follows by induction. (For an intuitive proof, see [Example 2.33](#).)

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_i A_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n P(A_i A_j A_k) \\ &\quad - \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n). \end{aligned}$$

Example 1.18 Suppose that 25% of the population of a city read newspaper A, 20% read newspaper B, 13% read C, 10% read both A and B, 8% read both A and C, 5% read B and C, and 4% read all three. If a person from this city is selected at random, what is the probability that he or she does not read any of these newspapers?

Solution: Let E , F , and G be the events that the person reads A, B, and C, respectively. The event that the person reads at least one of the newspapers A, B, or C is $E \cup F \cup G$. Therefore,

$1 - P(E \cup F \cup G)$ is the probability that he or she reads none of them. Since

$$\begin{aligned} P(E \cup F \cup G) &= P(E) + P(F) + P(G) - P(EF) - P(EG) \\ &\quad - P(FG) + P(EFG) \\ &= 0.25 + 0.20 + 0.13 - 0.10 - 0.08 - 0.05 + 0.04 = 0.39, \end{aligned}$$

the desired probability equals $1 - 0.39 = 0.61$. ♦

Example 1.19 Dr. Grossman, an internist, has 520 patients, of which (i) 230 are hypertensive, (ii) 185 are diabetic, (iii) 35 are hypochondriac and diabetic, (iv) 25 are all three, (v) 150 are none, (vi) 140 are only hypertensive, and finally, (vii) 15 are hypertensive and hypochondriac but not diabetic. Assume that appointments are all random. This implies that even hypochondriacs do not make more visits than others.

Solution: Let T , C , and D denote the events that the next appointment of Dr. Grossman is hypertensive, hypochondriac, and diabetic, respectively. The Venn diagram of Figure 1.6 shows that the number of patients with only hypochondria is 30. Therefore, the desired probability is $30/520 \approx 0.06$. ♦

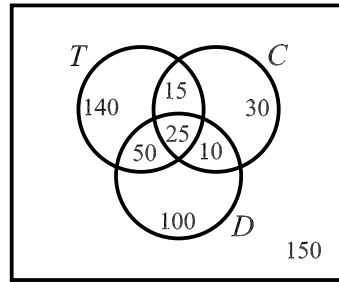


Figure 1.6 Venn diagram of Example 1.19.

Theorem 1.7 $P(A) = P(AB) + P(AB^c)$.

Proof: Clearly, $A = AS = A(B \cup B^c) = AB \cup AB^c$. Since AB and AB^c are mutually exclusive,

$$P(A) = P(AB \cup AB^c) = P(AB) + P(AB^c). \quad \blacklozenge$$

Example 1.20 In a community, 32% of the population are male smokers; 27% are female smokers. What percentage of the population of this community smoke?

Solution: Let A be the event that a randomly selected person from this community smokes. Let B be the event that the person is male. By Theorem 1.7,

$$P(A) = P(AB) + P(AB^c) = 0.32 + 0.27 = 0.59.$$

Therefore, 59% of the population of this community smoke. ♦

Remark 1.4 In the real world, especially for games of chance, it is common to express probability in terms of **odds**. We say that the *odds in favor of an event A are r to s* if $P(A) = r/(r + s)$. Similarly, the *odds against an event A are r to s* if $P(A) = s/(r + s)$. Therefore, if the odds in favor of an event A are r to s , then the odds against A are s to r . For example, in drawing a card at random from an ordinary deck of 52 cards, the odds against drawing an ace are 48 to 4 or, equivalently, 12 to 1. The odds in favor of an ace are 4 to 48 or, equivalently, 1 to 12. If for an event A , $P(A) = p$, then the odds in favor of A are p to $1 - p$. Therefore, for example, in flipping a fair coin three times, by [Example 1.12](#), the odds in favor of HHH are $1/8$ to $7/8$ or, equivalently, 1 to 7. ♦

EXERCISES

A

1. Gottfried Wilhelm Leibniz (1646–1716), the German mathematician, philosopher, statesman, and one of the supreme intellects of the seventeenth century, believed that in a throw of a pair of fair dice, the probability of obtaining the sum 11 is equal to that of obtaining the sum 12. Do you agree with Leibniz? Explain.
2. Show that if A and B are mutually exclusive, then $P(A) + P(B) \leq 1$.
3. Suppose that 33% of the people have O^+ blood and 7% have O^- . What is the probability that the next president of the United States has type O blood?
4. Sixty-eight minutes through *Morituri*, the 1965 Marlon Brando movie, a group of underground anti-Nazis wanting to escape from a Nazi ship to a nearby island state that the chances of their succeeding are 15-to-1. Eighty-two minutes through the movie, the same group refers to this estimate stating that the odds against successfully escaping to that island are 15-to-1. Are these two statements consistent? Why or why not?
5. The probability that an earthquake will damage a certain structure during a year is 0.015. The probability that a hurricane will damage the same structure during a year is 0.025. If the probability that *both* an earthquake and a hurricane will damage the structure during a year is 0.0073, what is the probability that next year the structure will not be damaged by a hurricane or an earthquake?
6. In a probability test, for two events E and F of a sample space, Tina's calculations resulted in $P(E) = 1/4$, $P(F) = 1/2$, and $P(EF) = 3/8$. Is it possible that Tina made a mistake in her calculations? Why or why not?
7. For events A and B , suppose that the probability that at least one of them occurs is 0.8 and the probability that both of them occur is 0.3. Find the probability that exactly one of them occurs.
8. The admission office of a college admits only applicants whose high school GPA is at least 3.0 or whose SAT score is 1200 or higher. If 38% of the applicants of this college have at least a 3.0 GPA, 30% have a SAT of 1200 or higher, and 15% have both, what percentage of all the applicants are admitted to the college?
9. Jacqueline, Bonnie, and Tina are the only contestants in an athletic race, where it is not possible to tie. The probability that Bonnie wins is $2/3$ that of Jacqueline winning and $4/3$ that of Tina winning. Find the probability of each of these three athletes winning.

10. A motorcycle insurance company has 7000 policyholders of whom 5000 are under 40. If 4100 of the policyholders are males and under 40, 1100 are married and under 40, and 550 are married males who are under 40, find the probability that the next motorcycle policyholder of this company who gets into an accident is a single female who is under 40. Assume that it is equally likely for all policyholders to get into an accident.
11. From an ordinary deck of 52 cards, we draw cards at random and without replacement. What is the probability that at least two cards must be drawn to obtain a face card?
12. Suppose that the probability that a driver is a male, and has at least one motor vehicle accident during a one-year period, is 0.12. Suppose that the corresponding probability for a female is 0.06. What is the probability of a randomly selected driver having at least one accident during the next 12 months?
13. Suppose that 75% of all investors invest in traditional annuities and 45% of them invest in the stock market. If 85% invest in the stock market and/or traditional annuities, what percentage invest in both?
14. In a horse race, the odds in favor of the first horse winning in an 8-horse race are 2 to 5. The odds against the second horse winning are 7 to 3. What is the probability that one of these horses will win?
15. A professor asks her students to present three methods of generating a random state out of US states. One of the methods a student, Walter, introduces is to draw a congressman from the list of 535 voting members of the House of Representatives and then select the state to which that member belongs. Is the state selected by this method random?
16. Suppose that in the following table, by the probability of the interval $[x, y)$ we mean the probability that, in a certain region, a person dies on or after his or her x th birthday but before his or her y th birthday.

Interval	Probability of the interval	Interval	Probability of the interval
$[0, 10)$	0.0634	$[50, 60)$	0.1257
$[10, 20)$	0.0129	$[60, 70)$	0.2089
$[20, 30)$	0.0218	$[70, 80)$	0.2710
$[30, 40)$	0.0317	$[80, 90)$	0.1683
$[40, 50)$	0.0634	$[90, 100)$	0.0277
		$[100, \infty)$	0.0052

Based on this table of mortality rates, what is the probability that a baby just born in this region reaches the age of 60? What is the probability that he or she lives to be at least 20 but dies before 50?

17. Let $S = \{\omega_1, \omega_2, \omega_3\}$ be the sample space of an experiment. If $P(\{\omega_1, \omega_2\}) = 0.5$ and $P(\{\omega_1, \omega_3\}) = 0.7$, find $P(\{\omega_1\})$, $P(\{\omega_2\})$, and $P(\{\omega_3\})$.
18. Excerpt from the TV show *The Rockford Files*:

Rockford: There are only two doctors in town. The chances of both autopsies being performed by the same doctor are 50–50.

Reporter: No, that is only for one autopsy. For two autopsies, the chances are 25–75.

Rockford: You're right.

Was Rockford right to agree with the reporter? Explain why or why not?

19. In a major state university, to study the performance relationship between calculus I and calculus II grades, the mathematics department reviewed the letter grades of all students who had taken both of the calculus courses in the last twenty years. For each pair of letter grades (X, Y) , the department calculated the proportion of all of the students who had earned X in calculus I and Y in calculus II. The results are given in the following table.

Calculus I	Calculus II				
	F	D	C	B	A
F	0.00	0.00	0.00	0.00	0.00
D	0.08	0.05	0.04	0.02	0.01
C	0.02	0.03	0.18	0.08	0.02
B	0.01	0.01	0.09	0.07	0.04
A	0.00	0.01	0.06	0.10	0.08

According to the rules of this university, if a student takes a course several times, only the last grade will be recorded. Based on this table, find the probability that a randomly selected student (a) does better in calculus I than in calculus II; (b) does better in calculus II than in calculus I; (c) earns the same letter grade in both calculus courses.

20. A company has only one position with three highly qualified applicants: John, Barbara, and Marty. However, because the company has only a few women employees, Barbara's chance to be hired is 20% higher than John's and 20% higher than Marty's. Find the probability that Barbara will be hired.
21. In a psychiatric hospital, the number of patients with schizophrenia is three times the number with psychoneurotic reactions, twice the number with alcohol addictions, and 10 times the number with involutional psychotic reaction. If a patient is selected randomly from the list of all patients with one of these four diseases, what is the probability that he or she suffers from schizophrenia? Assume that none of these patients has more than one of these four diseases.
22. Let A and B be two events. Prove that

$$P(AB) \geq P(A) + P(B) - 1.$$

23. A card is drawn at random from an ordinary deck of 52 cards. What is the probability that it is (a) a black ace or a red queen; (b) a face or a black card; (c) neither a heart nor a queen?
24. Which of the following statements is true? If a statement is true, prove it. If it is false, give a counterexample.
- (a) If $P(A) + P(B) + P(C) = 1$, then the events A, B , and C are mutually exclusive.
- (b) If $P(A \cup B \cup C) = 1$, then A, B , and C are mutually exclusive events.

25. Suppose that in the Baltimore metropolitan area 25% of the crimes occur during the day and 80% of the crimes occur in the city. If only 10% of the crimes occur outside the city during the day, what percent occur inside the city during the night? What percent occur outside the city during the night?
26. Suppose that, in a temperate coniferous forest, 40% of randomly selected quarter-acre plots have both cedar and cypress trees, 25% have both cedar and redwood trees, 20% have cypress and redwoods, and 15% have all three. What is the probability that in a randomly selected quarter-acre plot in this forest we can find exactly two of the three types of these trees?
27. Let A , B , and C be three events. Prove that
- $$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC).$$
28. Let A , B , and C be three events. Show that exactly two of these events will occur with probability
- $$P(AB) + P(AC) + P(BC) - 3P(ABC).$$
29. Eleven chairs are numbered 1 through 11. Four girls and seven boys sit on these chairs at random. What is the probability that chair 5 is occupied by a boy?
30. A ball is thrown at a square that is divided into n^2 identical squares. The probability that the ball hits the square of the i th column and j th row is p_{ij} , where $\sum_{i=1}^n \sum_{j=1}^n p_{ij} = 1$. In terms of p_{ij} 's, find the probability that the ball hits the j th horizontal strip.
31. Among 33 students in a class, 17 of them earned A 's on the midterm exam, 14 earned A 's on the final exam, and 11 did not earn A 's on either examination. What is the probability that a randomly selected student from this class earned an A on both exams?
32. From a small town 120 persons were selected at random and asked the following question: Which of the three shampoos, A , B , or C , do you use? The following results were obtained: 20 use A and C , 10 use A and B but not C , 15 use all three, 30 use only C , 35 use B but not C , 25 use B and C , and 10 use none of the three. If a person is selected at random from this group, what is the probability that he or she uses (a) only A ; (b) only B ; (c) A and B ? (Draw a Venn diagram.)
33. The coefficients of the quadratic equation $x^2 + bx + c = 0$ are determined by tossing a fair die twice (the first outcome is b , the second one is c). Find the probability that the equation has real roots.
34. Two integers m and n are called *relatively prime* if 1 is their only common positive divisor. Thus 8 and 5 are relatively prime, whereas 8 and 6 are not. A number is selected at random from the set $\{1, 2, 3, \dots, 63\}$. Find the probability that it is relatively prime to 63.
35. A number is selected randomly from the set $\{1, 2, \dots, 1000\}$. What is the probability that (a) it is divisible by 3 but not by 5; (b) it is divisible neither by 3 nor by 5?
36. The secretary of a college has calculated that from the students who took calculus, physics, and chemistry last semester, 78% passed calculus, 80% physics, 84% chemistry, 60% calculus and physics, 65% physics and chemistry, 70% calculus and chemistry, and 55% all three. Show that these numbers are not consistent, and therefore the secretary has made a mistake.

B

37. From an ordinary deck of 52 cards, we draw cards at random and without replacement until only cards of one suit are left. Find the probability that the cards left are all spades.
38. A number is selected at random from the set of natural numbers $\{1, 2, \dots, 1000\}$. What is the probability that it is divisible by 4 but neither by 5 nor by 7?
39. For a Democratic candidate to win an election, she must win districts I, II, and III. Polls have shown that the probability of winning I and III is 0.55, losing II but not I is 0.34, and losing II and III but not I is 0.15. Find the probability that this candidate will win all three districts. (Draw a Venn diagram.)
40. For events E and F , show that

$$P(E \cup F) + P(E \cup F^c) + P(E^c \cup F) + P(E^c \cup F^c) = 3.$$

41. Two numbers are successively selected at random and with replacement from the set $\{1, 2, \dots, 100\}$. What is the probability that the first one is greater than the second?
42. Let $A_1, A_2, A_3, \dots$ be a sequence of events of a sample space. Prove that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

This is called **Boole's inequality**.

43. Let $A_1, A_2, A_3, \dots$ be a sequence of events of an experiment. Prove that

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) \geq 1 - \sum_{n=1}^{\infty} P(A_n^c).$$

Hint: Use Boole's inequality, discussed in [Exercise 42](#).

44. In a certain country, the probability is 49/50 that a randomly selected fighter plane returns from a mission without mishap. Mia argues that this means there is one mission with a mishap in every 50 consecutive flights. She concludes that if a fighter pilot returns safely from 49 consecutive missions, he should return home before his fiftieth mission. Is Mia right? Explain why or why not.
45. Let P be a probability defined on a sample space S . For events A of S define $Q(A) = [P(A)]^2$ and $R(A) = P(A)/2$. Is Q a probability on S ? Is R a probability on S ? Why or why not?

In its general case, the following exercise has important applications in coding theory, telecommunications, and computer science.

46. **(The Hat Problem)** A game begins with a team of three players entering a room one at a time. For each player, a fair coin is tossed. If the outcome is heads, a red hat is placed on the player's head, and if it is tails, a blue hat is placed on the player's head. The players are allowed to communicate before the game begins to decide on a strategy. However, no communication is permitted after the game begins. Players cannot see their own hats. But each player can see the other two players' hats. Each player is

given the option to guess the color of his or her hat or to pass. The game ends when the three players simultaneously make their choices. The team wins if no player's guess is incorrect and at least one player's guess is correct. Obviously, the team's goal is to develop a strategy that maximizes the probability of winning. A trivial strategy for the team would be for two of its players to pass and the third player to guess red or blue as he or she wishes. This gives the team a 50% chance to win. Can you think of a strategy that improves the chances of the team winning?



Self-Quiz on Section 1.4

Time allotted: 20 Minutes

1. Let A and B be mutually exclusive events of an experiment with $P(A) = 1/3$ and $P(B) = 1/4$. What is the probability that neither A occurs nor B ? (3 points)
2. Zack has two weeks to read a book assigned by his English teacher and two weeks to write an essay assigned by his philosophy professor. The probability that he completes both assignments on time is 0.6, and the probability that he completes at least one of them on time is 0.95. What is the probability that he ends up meeting the deadline only for one assignment? (3 points)
3. Suppose that two-thirds of Americans traveling to Europe visit at least one of the three countries France, England, and Italy. Furthermore, suppose that one-half of them visit England, one-third visit France, and one-fourth visit Italy. If for each pair of these countries, one-fifth of the travelers visit that pair, what fraction of Americans traveling to Europe visit all of these three countries? (4 points)



1.5 CONTINUITY OF PROBABILITY FUNCTIONS

Let $\mathbf{R}$ denote (here and everywhere else throughout the book) the set of all real numbers. We know from calculus that a function $f: \mathbf{R} \rightarrow \mathbf{R}$ is called continuous at a point $c \in \mathbf{R}$ if $\lim_{x \rightarrow c} f(x) = f(c)$. It is called continuous on $\mathbf{R}$ if it is continuous at all points $c \in \mathbf{R}$. We also know that this definition is equivalent to the sequential criterion $f: \mathbf{R} \rightarrow \mathbf{R}$ is continuous on $\mathbf{R}$ if and only if, for every convergent sequence $\{x_n\}_{n=1}^{\infty}$ in $\mathbf{R}$,

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right). \quad (1.4)$$

This property, in some sense, is shared by the probability function. To explain this, we need to introduce some definitions. But first recall that probability is a set function from $\mathcal{P}(S)$, the set of all possible events of the sample space S , to $[0, 1]$.

A sequence $\{E_n, n \geq 1\}$ of events of a sample space is called **increasing** if

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \cdots ;$$

it is called **decreasing** if

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots \supseteq E_n \supseteq E_{n+1} \supseteq \cdots.$$

For an increasing sequence of events $\{E_n, n \geq 1\}$, by $\lim_{n \rightarrow \infty} E_n$ we mean the event that at least one $E_i, 1 \leq i < \infty$ occurs. Therefore,

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{i=1}^{\infty} E_i.$$

Similarly, for a decreasing sequence of events $\{E_n, n \geq 1\}$, by $\lim_{n \rightarrow \infty} E_n$ we mean the event that every E_i occurs. Thus in this case

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{i=1}^{\infty} E_i.$$

The following theorem expresses the property of probability function that is analogous to (1.4).

Theorem 1.8 (Continuity of Probability Function) *For any increasing or decreasing sequence of events, $\{E_n, n \geq 1\}$,*

$$\lim_{n \rightarrow \infty} P(E_n) = P(\lim_{n \rightarrow \infty} E_n).$$

Proof: For the case where $\{E_n, n \geq 1\}$ is increasing, let $F_1 = E_1, F_2 = E_2 - E_1, F_3 = E_3 - E_2, \dots, F_n = E_n - E_{n-1}, \dots$. Clearly, $\{F_i, i \geq 1\}$ is a mutually exclusive set of events that satisfies the following relations:

$$\begin{aligned} \bigcup_{i=1}^n F_i &= \bigcup_{i=1}^n E_i = E_n, & n = 1, 2, 3, \dots, \\ \bigcup_{i=1}^{\infty} F_i &= \bigcup_{i=1}^{\infty} E_i \end{aligned}$$

(see Figure 1.7). Hence

$$\begin{aligned} P(\lim_{n \rightarrow \infty} E_n) &= P\left(\bigcup_{i=1}^{\infty} E_i\right) = P\left(\bigcup_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} P(F_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(F_i) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n F_i\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n E_i\right) = \lim_{n \rightarrow \infty} P(E_n), \end{aligned}$$

where the last equality follows since $\{E_n, n \geq 1\}$ is increasing, and hence $\bigcup_{i=1}^n E_i = E_n$. This establishes the theorem for increasing sequences.

If $\{E_n, n \geq 1\}$ is decreasing, then $E_n \supseteq E_{n+1}, \forall n$, implies that $E_n^c \subseteq E_{n+1}^c, \forall n$. Therefore, the sequence $\{E_n^c, n \geq 1\}$ is increasing and

$$\begin{aligned} P(\lim_{n \rightarrow \infty} E_n) &= P\left(\bigcap_{i=1}^{\infty} E_i\right) = 1 - P\left[\left(\bigcap_{i=1}^{\infty} E_i\right)^c\right] = 1 - P\left(\bigcup_{i=1}^{\infty} E_i^c\right) \\ &= 1 - P(\lim_{n \rightarrow \infty} E_n^c) = 1 - \lim_{n \rightarrow \infty} P(E_n^c) = 1 - \lim_{n \rightarrow \infty} [1 - P(E_n)] \\ &= 1 - 1 + \lim_{n \rightarrow \infty} P(E_n) = \lim_{n \rightarrow \infty} P(E_n). \quad \blacklozenge \end{aligned}$$

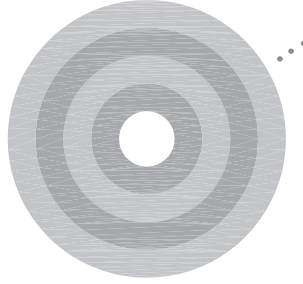


Figure 1.7 The circular disks are the E_i 's and the shaded circular annuli are the F_i 's, except for F_1 , which equals E_1 .

Example 1.21 Suppose that some individuals in a population produce offspring of the same kind. The offspring of the initial population are called second generation, the offspring of the second generation are called third generation, and so on. Furthermore, suppose that with probability $\exp[-(2n^2 + 7)/(6n^2)]$ the entire population completely dies out by the n th generation before producing any offspring, what is the probability that such a population survives forever?

Solution: Let E_n denote the event of extinction of the entire population by the n th generation; then

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \subseteq \cdots$$

because if E_n occurs, then E_{n+1} also occurs. Hence, by [Theorem 1.8](#),

$$\begin{aligned} P(\{\text{population survives forever}\}) &= 1 - P(\{\text{population eventually dies out}\}) \\ &= 1 - P\left(\bigcup_{i=1}^{\infty} E_i\right) = 1 - \lim_{n \rightarrow \infty} P(E_n) \\ &= 1 - \lim_{n \rightarrow \infty} \exp\left(-\frac{2n^2 + 7}{6n^2}\right) = 1 - e^{-1/3}. \quad \blacklozenge \end{aligned}$$

1.6 PROBABILITIES 0 AND 1

Events with probabilities 1 and 0 should not be misinterpreted. If E and F are events with probabilities 1 and 0, respectively, it is not correct to say that E is the sample space S and F is the empty set $\emptyset$. In fact, there are experiments in which there exist infinitely many events each with probability 1, and infinitely many events each with probability 0. An example follows.

Suppose that an experiment consists of selecting a random point from the interval $(0, 1)$. Since every point in $(0, 1)$ has a decimal representation such as

$$0.529387043219721 \cdots,$$

the experiment is equivalent to picking an endless decimal from $(0, 1)$ at random (note that if a decimal terminates, all of its digits from some point on are 0). In such an experiment we want to compute the probability of selecting the point $1/3$. In other words, we want to compute the probability of choosing $0.333333 \cdots$ in a random selection of an endless decimal. Let A_n be the event that the selected decimal has 3 as its first n digits; then

$$A_1 \supset A_2 \supset A_3 \supset A_4 \supset \cdots \supset A_n \supset A_{n+1} \supset \cdots,$$

since the occurrence of A_{n+1} guarantees the occurrence of A_n . Now $P(A_1) = 1/10$ because there are 10 choices 0, 1, 2, ..., 9 for the first digit, and we want only one of them, namely 3, to occur. $P(A_2) = 1/100$ since there are 100 choices 00, 01, ..., 09, 10, 11, ..., 19, 20, ..., 99 for the first two digits, and we want only one of them, 33, to occur. $P(A_3) = 1/1000$ because there are 1000 choices 000, 001, ..., 999 for the first three digits, and we want only one of them, 333, to occur. Continuing this argument, we have $P(A_n) = (1/10)^n$. Since $\bigcap_{n=1}^{\infty} A_n = \{1/3\}$, by [Theorem 1.8](#),

$$P\left(\frac{1}{3} \text{ is selected}\right) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \left(\frac{1}{10}\right)^n = 0.$$

Note that there is nothing special about the point $1/3$. For any other point $0.\alpha_1\alpha_2\alpha_3\alpha_4\cdots$ from $(0, 1)$, the same argument could be used to show that the probability of its occurrence is 0 (define A_n to be the event that the first n digits of the selected decimal are $\alpha_1, \alpha_2, \dots, \alpha_n$, respectively, and repeat the same argument). We have shown that in random selection of points from $(0, 1)$, the probability of the occurrence of any particular point is 0. Now for $t \in (0, 1)$, let $B_t = (0, 1) - \{t\}$. Then $P(\{t\}) = 0$ implies that

$$P(B_t) = P(\{t\}^c) = 1 - P(\{t\}) = 1.$$

Therefore, there are infinitely many events, B_t 's, each with probability 1 and none equal to the sample space $(0, 1)$.

1.7 RANDOM SELECTION OF POINTS FROM INTERVALS

In [Section 1.6](#), we showed that the probability of the occurrence of any particular point in a random selection of points from an interval (a, b) is 0. This implies immediately that if $[\alpha, \beta] \subseteq (a, b)$, then the events that the point falls in $[\alpha, \beta]$, (α, β) , $[\alpha, \beta)$, and $(\alpha, \beta]$ are all equiprobable. Now consider the intervals $\left(a, \frac{a+b}{2}\right)$ and $\left(\frac{a+b}{2}, b\right)$; since $\frac{a+b}{2}$ is the midpoint of (a, b) , it is reasonable to assume that

$$p_1 = p_2, \tag{1.5}$$

where p_1 is the probability that the point belongs to $\left(a, \frac{a+b}{2}\right)$ and p_2 is the probability that it belongs to $\left(\frac{a+b}{2}, b\right)$. The events that the random point belongs to $\left(a, \frac{a+b}{2}\right)$ and $\left(\frac{a+b}{2}, b\right)$ are mutually exclusive and

$$\left(a, \frac{a+b}{2}\right) \cup \left[\frac{a+b}{2}, b\right) = (a, b);$$

therefore,

$$p_1 + p_2 = 1.$$

This relation and [\(1.5\)](#) imply that

$$p_1 = p_2 = 1/2.$$

Hence the probability that a random point selected from (a, b) falls into the interval $\left(a, \frac{a+b}{2}\right)$ is $1/2$. The probability that it falls into $\left[\frac{a+b}{2}, b\right)$ is also $1/2$. Note that the length of each of these intervals is $1/2$ of the length of (a, b) . Now consider the intervals $\left(a, \frac{2a+b}{3}\right]$, $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$, and $\left(\frac{a+2b}{3}, b\right)$. Since $\frac{2a+b}{3}$ and $\frac{a+2b}{3}$ are the points that divide the interval (a, b) into three subintervals with equal lengths, we can assume that

$$p_1 = p_2 = p_3, \quad (1.6)$$

where p_1 , p_2 , and p_3 are the probabilities that the point falls into $\left(a, \frac{2a+b}{3}\right]$, $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$, and $\left(\frac{a+2b}{3}, b\right)$, respectively. On the other hand, these three intervals are mutually disjoint and

$$\left(a, \frac{2a+b}{3}\right] \cup \left(\frac{2a+b}{3}, \frac{a+2b}{3}\right] \cup \left(\frac{a+2b}{3}, b\right) = (a, b).$$

Hence

$$p_1 + p_2 + p_3 = 1.$$

This relation and (1.6) imply that

$$p_1 = p_2 = p_3 = 1/3.$$

Therefore, the probability that a random point selected from (a, b) falls into the interval $\left(a, \frac{2a+b}{3}\right]$ is $1/3$. The probability that it falls into $\left(\frac{2a+b}{3}, \frac{a+2b}{3}\right]$ is $1/3$, and the probability that it falls into $\left(\frac{a+2b}{3}, b\right)$ is $1/3$. Note that the length of each of these intervals is $1/3$ of the length of (a, b) . These and other similar observations indicate that the probability of the event that a random point from (a, b) falls into a subinterval (α, β) is equal to $(\beta - \alpha)/(b - a)$.

Note that in this discussion we have assumed that subintervals of equal lengths are equiprobable. Even though two subintervals of equal lengths may differ by a finite or a countably infinite set (or even a set of measure zero), this assumption is still consistent with our intuitive understanding of choosing random points from intervals. This is because in such an experiment, the probability of the occurrence of a finite or countably infinite set (or a set of measure zero) is 0.

Thus far, we have based our discussion of selecting random points from intervals on our intuitive understanding of this experiment and not on a mathematical definition. Such discussions are often necessary for the creation of appropriate mathematical meanings for unclear concepts. The following definition, which is based on our intuitive analysis, gives an exact mathematical meaning to the experiment of random selection of points from intervals.

Definition 1.2 A point is said to be **randomly selected from an interval** (a, b) if any two subintervals of (a, b) that have the same length are equally likely to include the point. The

probability associated with the event that the subinterval (α, β) contains the point is defined to be $(\beta - \alpha)/(b - a)$.

As explained before, choosing a random number from $(0, 1)$ is equivalent to choosing randomly all the decimal digits of the number successively. Since in practice this is impossible, choosing exact random points or numbers from $(0, 1)$ or any other interval is only a theoretical matter. Approximate random numbers, however, can be generated by computers. Most of the computer languages, some scientific computer software, and some calculators are equipped with subroutines that generate approximate random numbers from intervals. However, since it is difficult to construct good random number generators, there are computer languages, software programs, and calculators that are equipped with poor random number generator algorithms. An excellent reference for construction of good random number generators is *The Art of Computer Programming*, Volume 2, *Seminumerical Algorithms*, third edition, by Donald E. Knuth (Addison Wesley, 1998). Simple mechanical tools can also be used to find such approximations. For example, consider a spinner mounted on a wheel of *unit circumference* (radius $1/2\pi$). Let A be a point on the perimeter of the wheel. Each time that we flick the spinner, it stops, pointing toward some point B on the wheel's circumference. The length of the arc AB (directed, say, counterclockwise) is an approximate random number between 0 and 1 if the spinner does not have any “sticky” spots (see Figure 1.8).

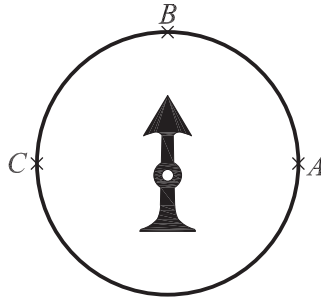


Figure 1.8 Spinner, a model to generate random numbers.

Finally, when selecting a random point from an interval (a, b) , we may think of an extremely large hypothetical box that contains infinitely many indistinguishable balls. Imagine that each ball is marked by a number from (a, b) , each number of (a, b) is marked on exactly one ball, and the balls are completely mixed up, so that in a random selection of balls, any two of them have the same chance of being drawn. With this transcendental model in mind, choosing a random number from (a, b) is then equivalent to drawing a random ball from such a box and looking at its number.

★ Example 1.22 (A Set that Is Not an Event) Let an experiment consist of selecting a point at random from the interval $[-1, 2]$. We will construct a set that is not an event (i.e., it is impossible to associate a probability with the set). We begin by defining an equivalence relation on $[0, 1]$: $x \sim y$ if $x - y$ is a rational number. Let $\mathbf{Q} = \{r_1, r_2, \dots\}$ be the set of rational numbers in $[-1, 1]$. Clearly, x is equivalent to y if $x - y \in \mathbf{Q}$. The fact that this relation is reflexive, symmetric, and transitive is trivial. Therefore, being an equivalence relation, it partitions the interval $[0, 1]$ into disjoint equivalence classes (Λ_α) . These classes are such that if x and $y \in \Lambda_\alpha$ for some α , then $x - y$ is rational. However, if $x \in \Lambda_\alpha$ and $y \in \Lambda_\beta$, and $\alpha \neq \beta$, then $x - y$ is irrational. These observations imply that for each α , the equivalence

class Λ_α is countable. Since $\bigcup_\alpha \Lambda_\alpha = [0, 1]$ is uncountable, the number of equivalence classes is uncountable. Let E be a set consisting of *exactly* one point from each equivalence class Λ_α . The existence of such a set is guaranteed by the Axiom of Choice. We will show, by contradiction, that E is not an event. Suppose that E is an event, and let p be the probability associated with E . For each positive integer n , let $E_n = \{r_n + x : x \in E\} \subseteq [-1, 2]$. For each $r_n \in \mathbf{Q}$, E_n is simply a translation of E . Thus, for all n , the set E_n is also an event, and $P(E_n) = P(E) = p$.

We now make two more observations: (1) For $n \neq m$, $E_n \cap E_m = \emptyset$, (2) $[0, 1] \subset \bigcup_{n=1}^{\infty} E_n$. To prove (1), let $t \in E_n \cap E_m$. We will show that $E_n = E_m$. If $t \in E_n \cap E_m$, then for some $r_n, r_m \in \mathbf{Q}$, and $x, y \in E$, we have that $t = r_n + x = r_m + y$. That is, $x - y = r_m - r_n$ is rational, and $x - y$ belongs to the same equivalence class. Since E has exactly one point from each equivalence class, we must have $x = y$, hence $r_n = r_m$, hence $E_n = E_m$. To prove (2), let $x \in [0, 1]$. Then $x \sim y$ for some $y \in E$. This implies that $x - y$ is a rational number in $\mathbf{Q}$. That is, for some n , $x - y = r_n$, or $x = y + r_n$, or $x \in E_n$. Thus $x \in \bigcup_{n=1}^{\infty} E_n$.

Putting (1) and (2) together, we obtain

$$1/3 = P([0, 1]) \leq P\left(\bigcup_{n=1}^{\infty} E_n\right) \leq 1,$$

or

$$1/3 \leq \sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} p \leq 1.$$

This is a contradiction because $\sum_{n=1}^{\infty} p$ is either 0 or ∞ . Hence E is not an event, and we cannot associate a probability with this set. ♦

EXERCISES

A

1. A bus arrives at a station every day at a random time between 1:00 P.M. and 1:30 P.M. What is the probability that a person arriving at this station at 1:00 P.M. will have to wait at least 10 minutes?
2. Past experience shows that every new book by a certain publisher captures randomly between 4 and 12% of the market. What is the probability that the next book by this publisher captures at most 6.35% of the market?
3. Which of the following statements are true? If a statement is true, prove it. If it is false, give a counterexample.
 - (a) If A is an event with probability 1, then A is the sample space.
 - (b) If B is an event with probability 0, then $B = \emptyset$.
4. Let A and B be two events. Show that if $P(A) = 1$ and $P(B) = 1$, then $P(AB) = 1$.
5. A point is selected at random from the interval $(0, 2000)$. What is the probability that it is an integer?

6. Suppose that a point is randomly selected from the interval $(0, 1)$. Using [Definition 1.2](#), show that all numerals are equally likely to appear as the first digit of the decimal representation of the selected point.
7. For an experiment with sample space $S = (0, 2)$, for $n \geq 1$, let $E_n = \left(0, 1 + \frac{1}{n}\right)$ and $P(E_n) = (3n + 2)/5n$. Find the probability of $E = (0, 1]$. Note that this is *not* the experiment of choosing a point at random from the interval $(0, 2)$ as defined in [Section 1.7](#).
8. A point is chosen at random from the interval $(-1, 1)$. Let E_1 be the event that it falls in the interval $(-1, 1/3]$, E_2 be the event that it falls in the interval $(-1, 1/9]$, E_3 be the event that it falls in the interval $(-1, 1/27]$ and, in general, for $1 \leq i < \infty$, E_i be the event that the point is in the interval $(-1, 1/3^i]$. Find $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$.
9. For the experiment of choosing a point at random from the interval $(0, 1)$, let $E_n = (1/2 - 1/2n, 1/2 + 1/2n)$, $n \geq 1$.
 - (a) Prove that $\bigcap_{n=1}^{\infty} E_n = \{1/2\}$.
 - (b) Using part (a) and the continuity of probability function, show that the probability of selecting $1/2$ in a random selection of a point from $(0, 1)$ is 0.

B

10. Is it possible to define a probability on a countably infinite sample space so that the outcomes are equally probable?
11. Let $A_1, A_2, \dots, A_n$ be n events. Show that if

$$P(A_1) = P(A_2) = \dots = P(A_n) = 1,$$

then $P(A_1 A_2 \dots A_n) = 1$.

12. A point is selected at random from the interval $(0, 1)$. What is the probability that it is rational? What is the probability that it is irrational?
13. Suppose that a point is randomly selected from the interval $(0, 1)$. Using [Definition 1.2](#), show that all numerals are equally likely to appear as the n th digit of the decimal representation of the selected point.
14. Let $\{A_1, A_2, A_3, \dots\}$ be a sequence of events. Prove that if the series $\sum_{n=1}^{\infty} P(A_n)$ converges, then $P\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n\right) = 0$. This is called the **Borel–Cantelli lemma**. It says that if $\sum_{n=1}^{\infty} P(A_n) < \infty$, the probability that infinitely many of the A_n 's occur is 0.
Hint: Let $B_m = \bigcup_{n=m}^{\infty} A_n$ and apply [Theorem 1.8](#) to $\{B_m, m \geq 1\}$.
15. Show that the result of [Exercise 11](#) is not true for an infinite number of events. That is, show that if $\{E_t: 0 < t < 1\}$ is a collection of events for which $P(E_t) = 1$, it is not necessarily true that $P\left(\bigcap_{t \in (0,1)} E_t\right) = 1$.

16. Let A be the set of rational numbers in $(0, 1)$. Since A is countable, it can be written as a sequence (i.e., $A = \{r_n : n = 1, 2, 3, \dots\}$). Prove that for any $\varepsilon > 0$, A can be covered by a sequence of open balls whose total length is less than ε . That is, $\forall \varepsilon > 0$, there exists a sequence of open intervals (α_n, β_n) such that $r_n \in (\alpha_n, \beta_n)$ and $\sum_{n=1}^{\infty} (\beta_n - \alpha_n) < \varepsilon$. This important result explains why in a random selection of points from $(0, 1)$ the probability of choosing a rational is zero.

Hint: Let $\alpha_n = r_n - \varepsilon/2^{n+2}$, $\beta_n = r_n + \varepsilon/2^{n+2}$.



Self-Quiz on Section 1.7

Time allotted: 20 Minutes

Each problem is worth 5 points.

- Suppose that for events A and B , $P(AB) = 0$. Does this imply that A and B are mutually exclusive? Why or why not?
- For the experiment of choosing a point at random from the interval $[0, 1]$, let $E_n = \left[\frac{1}{3} - \frac{1}{n+2}, \frac{1}{3} + \frac{2}{n+2}\right]$, $n \geq 1$. Applying the Continuity of Probability Function to E_n 's, show that $P(1/3 \text{ is selected}) = 0$.



1.8 WHAT IS SIMULATION?

Solving a scientific or an industrial problem usually involves mathematical analysis and/or simulation. To perform a simulation, we repeat an experiment a large number of times to assess the probability of an event or condition occurring. For example, to estimate the probability of at least one 6 occurring within four rolls of a die, we may do a large number of experiments rolling a die four times and calculate the number of times that at least one 6 is obtained. Similarly, to estimate the fraction of time that, in a certain bank all the tellers are busy, we may measure the lengths of such time intervals over a long period X , add them, and then divide by X . Clearly, in simulations, the key to reliable answers is *to perform the experiment a large number of times or over a long period of time, whichever is applicable*. Since manually this is almost impossible, simulations are carried out by computers. Only computers can handle millions of operations in short periods of time.

To simulate a problem that involves random phenomena, generating random numbers from the interval $(0, 1)$ is essential. In almost every simulation of a probabilistic model, we will need to select random points from the interval $(0, 1)$. For example, to simulate the experiment of tossing a fair coin, we draw a random number from $(0, 1)$. If it is in $(0, 1/2)$, we say that the outcome is heads, and if it is in $[1/2, 1)$, we say that it is tails. Similarly, in the simulation of die tossing, the outcomes 1, 2, 3, 4, 5, and 6, respectively, correspond to the events that the random point from $(0, 1)$ is in $(0, 1/6)$, $[1/6, 1/3)$, $[1/3, 1/2)$, $[1/2, 2/3)$, $[2/3, 5/6)$, and $[5/6, 1)$.

As discussed in [Section 1.7](#), choosing a random number from a given interval is, in practice, impossible. In real-world problems, to perform simulation we use **pseudorandom numbers** instead. To generate n pseudorandom numbers from a uniform distribution on an interval (a, b) , we take an initial value $x_0 \in (a, b)$, called the **seed**, and construct a function ψ so that the sequence $\{x_1, x_2, \dots, x_n\} \subset (a, b)$ obtained recursively from

$$x_{i+1} = \psi(x_i), \quad 0 \leq i \leq n-1, \quad (1.7)$$

satisfies certain statistical tests for randomness. (Choosing the tests and constructing the function ψ are complicated matters beyond the scope of this book.) The function ψ takes a seed and generates a sequence of pseudorandom numbers in the interval (a, b) . Clearly, in any pseudorandom number generating process, the numbers generated are rounded to a certain number of decimal places. Therefore, ψ can only generate a finite number of pseudorandom numbers, which implies that, eventually, some x_j will be generated a second time. From that point on, by (1.7), a pitfall is that the same sequence of numbers that appeared after x_j 's first appearance will reappear. Beyond that point, numbers are not effectively random. One important aspect of the construction of ψ is that the second appearance of any of the x_j 's is postponed as long as possible.

It should be noted that passing certain statistical tests for randomness does not mean that the sequence $\{x_1, x_2, \dots, x_n\}$ is a randomly selected sequence from (a, b) in the true mathematical sense discussed in [Section 1.7](#). It is quite surprising that there are deterministic real-valued functions ψ that, for each i , generate an x_{i+1} that is completely determined by x_i , and yet the sequence $\{x_1, x_2, \dots, x_n\}$ passes certain statistical tests for randomness.

For convenience, throughout this chapter, in practical problems, by random number we simply mean pseudorandom number. In general, for good choices of ψ , the generated numbers are usually sufficiently random for practical purposes.

Most of the computer languages and some scientific computer software are equipped with subroutines that generate random numbers from intervals and from sets of integers. For example, in *Mathematica*, from Wolfram Research, Inc. (<http://www.wolfram.com>), the command **Random[]** selects a random number from $(0, 1)$, **Random[Real, {a, b}]** chooses a random number from the interval (a, b) , and **Random[Integer, {m, m + n}]** picks up an integer from $\{m, m + 1, m + 2, \dots, m + n\}$ randomly. However, there are computer languages that are not equipped with a subroutine that generates random numbers, and there are a few that are equipped with poor algorithms. As mentioned in [Section 1.7](#), even though it is difficult to construct good pseudorandom number generators, an excellent reference for construction of such numbers is *The Art of Computer Programming*, Volume 2, 3rd edition, by Donald E. Knuth.

At this point, let us emphasize that the main goal of scientists and engineers is always to solve a problem mathematically. It is a mathematical solution that is accurate, exact, and completely reliable. Simulations cannot take the place of a rigorous mathematical solution. They are widely used (a) to find good estimations for solutions of problems that either cannot be modeled mathematically or whose mathematical models are too difficult to solve; (b) to get a better understanding of the behavior of a complicated phenomenon; and/or (c) to obtain a mathematical solution by acquiring insight into the nature of the problem, its functions, and the magnitude and characteristics of its solution. Intuitively, it is clear why the results that are obtained by simulations are good. Theoretically, most of them can be justified by the strong law of large numbers, which will be discussed in [Section 11.4](#).

[Chapter 13](#) of the *Companion Website* for this book concerns computer simulation. That chapter is divided into several sections, presenting algorithms that are used to find approximate solutions to complicated probabilistic problems. These sections can be studied independently

when relevant materials from earlier chapters are being studied, or they can be learned concurrently.

CHAPTER 1 SUMMARY

► If the outcome of an experiment is not certain but all of its possible outcomes are predictable in advance, then the set of all these possible outcomes is called the *sample space* of the experiment and is usually denoted by S . These outcomes are sometimes called *sample points*, or simply *points*, of the sample space. Certain subsets of S are referred to as *events*. So events are sets of points of the sample space. If the outcome of an experiment belongs to an event E , we say that the event E has *occurred*.

► In the study of probability theory the relations between different events of an experiment play a central role. The following are among the most important examples:

De Morgan's first law:

$$(E \cup F)^c = E^c F^c, \quad \left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c, \quad \left(\bigcup_{i=1}^{\infty} E_i \right)^c = \bigcap_{i=1}^{\infty} E_i^c.$$

De Morgan's second law:

$$(EF)^c = E^c \cup F^c, \quad \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c, \quad \left(\bigcap_{i=1}^{\infty} E_i \right)^c = \bigcup_{i=1}^{\infty} E_i^c.$$

Another useful relation between E and F , two arbitrary events of a sample space S , is

$$E = EF \cup EF^c.$$

► If the joint occurrence of two events E and F is impossible, we say that E and F are *mutually exclusive*. So E and F are mutually exclusive if $EF = \emptyset$. A set of events $\{E_1, E_2, \dots\}$ is called *mutually exclusive* if the joint occurrence of any two of them is impossible, that is, if every pair of them is mutually exclusive.

► **Probability Axioms:** Let S be the sample space of a random phenomenon. Suppose that to each event A of S , a number denoted by $P(A)$ is associated with A . If P satisfies the following axioms, then it is called a *probability* and the number $P(A)$ is said to be the *probability of A* .

Axiom 1 $P(A) \geq 0$.

Axiom 2 $P(S) = 1$.

Axiom 3 If $\{A_1, A_2, A_3, \dots\}$ is a sequence of mutually exclusive events (i.e., the joint occurrence of every pair of them is impossible: $A_i A_j = \emptyset$ when $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Some immediate implications of the axioms of probability are the following:

- $P(\emptyset) = 0$.
- Let $\{A_1, A_2, \dots, A_n\}$ be a mutually exclusive set of events. Then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

- Let S be the sample space of an experiment. If S has N points that are all equally likely to occur, then for any event A of S , $P(A) = N(A)/N$, where $N(A)$ is the number of points of A .
- For any event A , $P(A^c) = 1 - P(A)$.
- If $A \subseteq B$, then $P(B - A) = P(BA^c) = P(B) - P(A)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.
- $P(A \cup B) = P(A) + P(B) - P(AB)$.
- $P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1A_2) - P(A_1A_3) - P(A_2A_3) + P(A_1A_2A_3)$.
- **Inclusion-Exclusion Principle** To calculate $P(A_1 \cup A_2 \cup \dots \cup A_n)$, first find all of the possible intersections of events from $A_1, A_2, \dots, A_n$ and calculate their probabilities. Then add the probabilities of those intersections that are formed of an odd number of events, and subtract the probabilities of those formed of an even number of events. The following formula is an expression for this principle.

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) = & \sum_{i=1}^n P(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n P(A_iA_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n P(A_iA_jA_k) \\ & - \dots + (-1)^{n-1} P(A_1A_2 \dots A_n). \end{aligned}$$

- $P(A) = P(AB) + P(AB^c)$.
- **(Continuity of Probability Function)** For any increasing or decreasing sequence of events, $\{E_n, n \geq 1\}$,

$$\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right).$$

► A point is said to be *randomly selected from an interval* (a, b) if any two subintervals of (a, b) that have the same length are equally likely to include the point. The probability associated with the event that the subinterval (α, β) contains the point is defined to be $(\beta - \alpha)/(b - a)$.

REVIEW PROBLEMS

1. The number of minutes it takes for a certain animal to react to a certain stimulus is a random number between 2 and 4.3. Find the probability that the reaction time of such an animal to this stimulus is no longer than 3.25 minutes.
2. Two dice are rolled. What is the event that the outcomes are consecutive?
3. From a phone book, a phone number is selected at random. (a) What is the event that the last digit is an odd number? (b) What is the event that the last digit is divisible by 3?
4. Let $\mathcal{P}$ be the set of all subsets of $A = \{1, 2\}$. We choose two distinct sets randomly from $\mathcal{P}$. Define a sample space for this experiment, and describe the following events:
 - (a) The intersection of the sets chosen at random is empty.
 - (b) The sets are complements of each other.
 - (c) One of the sets contains more elements than the other.
5. In a tutoring center, there are three computers that can be up or down at any given time. For $1 \leq i \leq 3$, let E_i be the event that computer i is up at a random time. In terms of E_i 's, describe the event that at least two computers are up.
6. Aiden just bought a stock for \$320. Define a sample space for the price of this stock in two years. Define the event that he makes money selling this stock at that time.
7. In a certain experiment, whenever the event A occurs, the event B also occurs. Which of the following statements is true and why?
 - (a) If we know that A has not occurred, we can be sure that B has not occurred as well.
 - (b) If we know that B has not occurred, we can be sure that A has not occurred as well.
8. A department store accepts only its own credit card or an American Express card. Customers not carrying one of these two cards must pay with cash. If 47% of the customers of this store carry American Express, 32% carry the store's credit card, and 12% carry both, what percentage of the store customers have no choice but to pay with cash?
9. Kayla has two cars, an Audi and a Jeep. At a given time, whether these cars are operative or totaled, due to accidents, is of concern to her insurance company. Define a sample space for the driving conditions of these cars at a random time. What is the event that at least one of the two cars is operative at that random time? Assume that a car that is not totaled is operative.
10. For a saw blade manufacturer's products, the global demand, per month, for band saws is between 30 and 36 thousands; for reciprocating saws, it is between 28 and 33 thousands; for hole saws, it is between 300 and 600 thousands; and for hacksaws, it is between 500 and 650 thousands. Define a sample space for the demands for these four types of saws by this manufacturer in a random month. Describe the event that, in such a month, the demand for reciprocating saws is no more than 28.5 thousands and for hacksaws, it is

at least 570 thousands. In this exercise, by “between two numbers a and b ”, we mean inclusive of both a and b .

11. The following relations *are not* always true. In each case give an example to refute them.

(a) $P(A \cup B) = P(A) + P(B)$.

(b) $P(AB) = P(A)P(B)$.

12. A coin is tossed until, for the first time, the same result appears twice in succession. Define a sample space for this experiment.
13. In a midwest town, 80% of households have cable TV, 60% have an internet subscription, and 90% have at least one of these. What percentage of the households of this town have both cable TV and an internet subscription?
14. Let A and B be two events of an experiment with $P(A) = 1/3$ and $P(B) = 1/4$. Find the maximum value for $P(A \cup B)$.
15. The number of the patients now in a hospital is 63. Of these 37 are male and 20 are for surgery. If among those who are for surgery 12 are male, how many of the 63 patients are neither male nor for surgery?
16. Let A , B , and C be three events. Prove that

$$P(A \cup B \cup C) \leq P(A) + P(B) + P(C).$$

17. Let A , B , and C be three events. Show that

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

if and only if $P(AB) = P(AC) = P(BC) = 0$.

18. Suppose that 40% of the people in a community drink or serve white wine, 50% drink or serve red wine, and 70% drink or serve red or white wine. What percentage of the people in this community drink or serve both red and white wine?
19. Suppose that, in a temperate coniferous forest, 60% of randomly selected quarter-acre plots have cedar trees, 45% have cypress trees, 30% have redwoods, 40% have both cedar and cypress, 25% have cedar and redwoods, 20% have cypress and redwoods, and 80% of such plots have at least one of these trees. What is the probability that in a randomly selected quarter-acre plot in this forest we can find all three types of these trees?
20. Suppose that $P(E \cup F) = 0.75$ and $P(E \cup F^c) = 0.85$. Find $P(E)$.
21. Anthony, Bob, and Carl, three American race car drivers, will compete in a professional Trans-Am road race. Past records show that the probability that one of these former champions wins is $10/13$. If Anthony is twice as likely to win as Bob, and Bob is three times more likely than Carl to win the race, find the respective probabilities of these three people winning.
22. Five customers enter a wireless corporate store to buy smartphones. If the probability that at least two of them purchase an Android smartphone is 0.6 and the probability that all of them buy non-Android smartphones is 0.17, what is the probability that exactly one Android smartphone is purchased?

23. Answer the following question, asked of Marilyn Vos Savant in the “Ask Marilyn” column of *Parade Magazine*, March 3, 1996.

My dad heard this story on the radio. At Duke University, two students had received A's in chemistry all semester. But on the night before the final exam, they were partying in another state and didn't get back to Duke until it was over. Their excuse to the professor was that they had a flat tire, and they asked if they could take a make-up test. The professor agreed, wrote out a test and sent the two to separate rooms to take it. The first question (on one side of the paper) was worth 5 points, and they answered it easily. Then they flipped the paper over and found the second question, worth 95 points: ‘Which tire was it?’ What was the probability that both students would say the same thing? My dad and I think it's 1 in 16. Is that right?

24. Let A and B be two events. Suppose that $P(A)$, $P(B)$, and $P(AB)$ are given. What is the probability that neither A nor B will occur?
25. Let A and B be two events. The event $(A - B) \cup (B - A)$ is called the **symmetric difference** of A and B and is denoted by $A \Delta B$. Clearly, $A \Delta B$ is the event that exactly one of the two events A and B occurs. Show that

$$P(A \Delta B) = P(A) + P(B) - 2P(AB).$$

26. Let $S = \{\omega_1, \omega_2, \omega_3, \dots\}$ be the sample space of an experiment. Suppose that $P(\{\omega_1\}) = 1/8$ and, for a constant k , $0 < k < 1$, $P(\{\omega_{i+1}\}) = k \cdot P(\{\omega_i\})$ for $i \geq 1$. Find $P(\{\omega_i\})$, for $i > 1$.
27. A bookstore receives six boxes of books per month on six random days of each month. Suppose that two of those boxes are from one publisher, two from another publisher, and the remaining two from a third publisher. Define a sample space for the possible orders in which the boxes are received in a given month by the bookstore. Describe the event that the last two boxes of books received last month are from the same publisher.
28. Mildred, a commuter student studying at Western New England University, reported to her advisor that there are three traffic lights that she needs to pass while driving from home to school. She said that, based on her experience, 10% of the time all of the three traffic lights are green, 35% of the time exactly two of them are green, and 43% of the time exactly one of them is green. Assuming that Mildred's observations are correct, what is the probability that on a random day she encounters (a) no green traffic lights; (b) at most one non-green (red or yellow) light; (c) at least two non-green lights?
29. Suppose that in a certain town the number of people with blood type O and blood type A are approximately the same. The number of people with blood type B is 1/10 of those with blood type A and twice the number of those with blood type AB. Find the probability that the next baby born in the town has blood type AB.
30. A number is selected at random from the set of natural numbers $\{1, 2, 3, \dots, 1000\}$. What is the probability that it is not divisible by 4, 7, or 9?
31. A number is selected at random from the set $\{1, 2, 3, \dots, 150\}$. What is the probability that it is relatively prime to 150? See [Exercise 34](#), [Section 1.4](#), for the definition of relatively prime numbers.

32. A point is selected randomly from the interval $(0, 2)$. For $n \geq 2$, let E_n be the event that it is in the interval $(0, \sqrt[n]{3})$. Determine the event $\bigcap_{n=2}^{\infty} E_n$.
33. Suppose that each day the price of a stock moves up $1/8$ of a point, moves down $1/8$ of a point, or remains unchanged. For $i \geq 1$, let U_i and D_i be the events that the price of the stock moves up and down on the i th trading day, respectively. In terms of U_i 's and D_i 's, find an expression for the event that the price of the stock
- remains unchanged on the i th trading day;
 - moves up every day of the next n trading days;
 - remains unchanged on at least one of the next n trading days;
 - is the same as today after three trading days;
 - does not move down on any of the next n trading days.
34. A bus traveling from Baltimore to New York breaks down at a random location. What is the probability that the breakdown occurred after passing through Philadelphia? The distances from New York and Philadelphia to Baltimore are, respectively, 199 and 96 miles.
35. The coefficients of the quadratic equation $ax^2 + bx + c = 0$ are determined by tossing a fair die three times (the first outcome is a , the second one b , and the third one c). Find the probability that the equation has no real roots.



Self-Test on Chapter 1

Time allotted: 120 Minutes

Each problem is worth 10 points.

- To determine who pays for dinner, Crispin, Allison, and Terry each flip an unbiased coin. The one whose flip has a different face up will pay. If all of the flips land on the same face, they start all over again. (a) What is the probability that Crispin ends up paying for dinner; (b) what is the probability that no more than one round of flips is necessary?
- Every day, a major ball manufacturer produces at least 800, but no more than 1300 of each of its five products: baseballs, tennis balls, softballs, basketballs, and soccer balls. Define a sample space for the production levels of these five types of balls that this manufacturer manufactures on a random working day. Describe the event that, on this day, the difference between the numbers of baseballs and soccer balls produced will not exceed 100.
Hint: Let x_1, x_2, x_3, x_4 , and x_5 be the production level, in hundreds, for baseballs, tennis balls, softballs, basketballs, and soccer balls, respectively.
- A device that has three components fails if at least one of its components breaks down. The device is observed at a random time. Let A_i , $1 \leq i \leq 3$, denote the outcome that the i th component is operative at the random time. In terms of A_i 's, (a) define a sample space for the status of the system at the random time; (b) describe the event that the device is not operative at that random time.

4. Suppose that, in a particular geographical area, of people aged 55 and older, 13.4% suffer from Parkinson's disease, 11.3% suffer from Alzheimer's disease, and 2.26% suffer from both of these completely separate neurodegenerative illnesses. What percentage of the people from this age group in this geographic area have neither Parkinson's disease nor Alzheimer's disease?
5. For an experiment, E and F are two events with $P(E) = 0.4$ and $P(E^c F^c) = 0.35$. Calculate $P(E^c F)$.
6. Last semester, all freshman students of a college took calculus, biology, and English. If 18% of them received an A in calculus, 10% received an A in both calculus and biology, 13% received an A in both English and calculus, and 7% received an A in all three courses, what is the probability that a randomly selected freshman from this college, last semester, received an A in calculus, but not in biology and not in English?
7. For an experiment with sample space $S = (0, 2)$, for $n \geq 1$, let $E_n = (1 - 1/n, 1 + 1/n)$ and $P(E_n) = (2n + 1)/3n$. For this experiment, find the probability that the event $\{1\}$ occurs. Note that this is *not* the experiment of choosing a point at random from the interval $(0, 2)$, as defined in [Section 1.7](#).
8. Five customers enter a wireless corporate store to purchase smartphones. If the probability that at least three of them purchase an Android smartphone is 0.54, what is the probability that at most two of them buy such a phone?
Hint: For $0 \leq i \leq 5$, let A_i be the event that exactly i of these customers buy an Android smartphone. Note that $\sum_{i=0}^5 P(A_i) = 1$.
9. In some towns of a country, the harmful substances lead and asbestos fibers find their way into drinking water. In a study, it was found that, in 13% of those towns, the drinking water supplies have neither lead nor asbestos fibers, in 32% of them the drinking water supplies have lead, and in 43% the drinking water supplies have asbestos fibers. In what percentage of the towns are the drinking water supplies contaminated with exactly one of these two impurities?
10. Suppose that on a certain week, only one-third of the travelers visiting Paris took a trip to the Eiffel Tower's top, one-half visited the Louvre Museum, and one-third took a tour of Notre Dame Cathedral. If none of the tourists visited all three sites, and for each pair of the sites, one-fourth of the travelers visited that pair, find the fraction of this particular group of tourists who did not visit any of the three sites.

